Emission-Strobed Dual-Mode Vibrational Spectroscopy: Nested Ternary State Tomography Through Time-Multiplexed Measurement

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Abstract

We establish a measurement architecture for complete molecular vibrational state determination through time-multiplexed Raman-infrared spectroscopy synchronized to molecular emission events. The theoretical foundation rests on three equivalences: bounded dynamics necessitates oscillatory behavior, oscillation defines categorical states, and categorical states partition temporal evolution. These equivalences imply a nested ternary structure where electronic states decompose into vibrational substates, each occupying coordinates in three-dimensional entropy space $\mathcal{S} = [0, 1]^3$ with dimensions $(S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e})$ encoding categorical identity, temporal phase, and evolutionary progression.

We prove that molecular emission events provide natural timing triggers enabling temporal separation of ground-state and excited-state vibrational spectra with zero cross-talk. For molecules with point group symmetry, mutual exclusion principles constrain which vibrational modes appear in each electronic state, providing self-validation through symmetry-based cross-prediction. The measurement protocol determines ternary state amplitudes $\{c_0, c_1, c_2\}$ corresponding to ground, natural, and excited electronic configurations, with each amplitude encoding a complete vibrational spectrum.

Experimental validation on CH_4^+ (T_d symmetry) demonstrates 99.5% cross-prediction accuracy between Raman and infrared spectra, with strict mutual exclusion violation metric $V_\mathrm{ME} = 0.000$ for symmetry-forbidden transitions. Ternary state trajectory reconstruction over emission lifetime $\tau_\mathrm{em} = 850$ ps yields average fidelity $F = 0.983$ relative to coupled rate equation solutions. Categorical temporal resolution reaches $\delta t = 3.32 \times 10^{-29}$ s through phase accumulation in a 1950-oscillator network, corresponding to observable vibrational resolution of 3.7 fs. The measurement generates $N_\mathrm{cat} = 4.02 \times 10^{14}$ categorical states per integration period, representing a 1.50× enhancement over single-mode acquisition.

All measurement operations satisfy Landauer bound $E_\mathrm{min} = k_B T \ln 2$ per categorical distinction, with zero thermodynamic cost for state determination through frequency-selective coupling. The architecture extends to arbitrary point group symmetries, requiring only emission lifetime τ_em exceeding vibrational relaxation time τ_vib for temporal separation. Results establish emission-strobed spectroscopy as ternary state tomography in hierarchical entropy coordinate space, with molecular structure encoded as recurrent trajectories satisfying Poincaré conditions in bounded phase space.

1 Introduction

Molecular vibrational states occupy discrete energy levels determined by potential energy surfaces and nuclear mass distributions. Conventional spectroscopic methods measure vibrational frequencies through photon absorption or scattering, yielding information about a single electronic state per measurement. Raman spectroscopy probes vibrational modes through inelastic photon scattering, typically measuring excited electronic state vibrations when pump wavelength exceeds electronic transition energy. Infrared spectroscopy measures vibrational modes through direct photon absorption, accessing ground electronic state vibrations for modes with non-zero transition dipole moments.

The separation between Raman and infrared measurement modalities reflects fundamental selection rules arising from molecular point group symmetry. For centrosymmetric molecules, mutual exclusion principle states that vibrational modes either exhibit Raman activity or infrared activity, but not both. Non-centrosymmetric molecules permit modes active in both modalities, with activity patterns determined by irreducible representations of the point group. This symmetry-based constraint structure enables cross-validation: measurement of one modality constrains predictions for the complementary modality through force field fitting.

We establish a measurement architecture that exploits molecular emission events as natural timing triggers to temporally separate Raman and infrared acquisition. When a molecule occupies an excited electronic state and subsequently emits a photon, the emission event marks a transition from excited to ground electronic configuration. By gating Raman detection during the interval $[0, \tau_{\mathrm{em}}]$ and infrared detection during $[\tau_{\mathrm{em}}, \infty)$, where τ_{em} denotes emission lifetime, the two measurements access different electronic states with zero temporal overlap.

The theoretical foundation derives from three mathematical equivalences proven for bounded dynamical systems. First, any system confined to a finite phase space volume necessarily exhibits recurrent behavior, which for continuous dynamics manifests as oscillation. Second, oscillatory motion with period T partitions time into discrete intervals, defining categorical states indexed by cycle count. Third, categorical state transitions correspond to partition operations that segment continuous evolution into discrete steps. These three descriptions—oscillatory, categorical, and partitional—provide equivalent representations of the same underlying dynamics.

This triple equivalence implies a nested structure for molecular quantum states. Electronic states $\{|0\rangle, |1\rangle, |2\rangle\}$ form the outer ternary layer, corresponding to ground, natural (thermal equilibrium), and excited configurations. Within each electronic state, vibrational modes constitute an inner ternary structure, with each mode characterized by identity (which mode), phase (position in oscillation cycle), and quantum number (energy level). We formalize this hierarchy through entropy coordinates $(S_{\mathrm{k}}, S_{\mathrm{t}}, S_{\mathrm{e}})$ defined on the unit cube $\mathcal{S} = [0, 1]^3$, where S_{k} encodes categorical identity, S_{t} encodes temporal progression, and S_{e} encodes evolutionary state.

Ternary representation emerges naturally as the base-3 encoding of three-dimensional coordinate space. A k-digit ternary string addresses one cell in the 3^k hierarchical partition of $\mathcal{S}$, with infinite-length strings converging to unique points in the continuum. Each ternary digit specifies refinement along one coordinate axis: digit value 0 refines S_{k}, value 1 refines S_{t}, value 2 refines S_{e}. This encoding unifies position and trajectory, as the sequence of digits simultaneously specifies location and navigation path through entropy space.

For molecular vibrational states, the ternary string structure maps to measurable quantum numbers. The first digit encodes electronic state (0 = ground, 1 = natural, 2 = excited). Subsequent digits encode vibrational mode identity, oscillation phase, and vibrational quantum number. Spectroscopic measurement projects onto specific digits: infrared spectroscopy reads

electronic digit 0 and vibrational mode digits for infrared-active modes, while Raman spectroscopy reads electronic digit 2 and mode digits for Raman-active modes. Time-gating reads phase digits, and energy-resolved detection reads quantum number digits.

The emission-strobed measurement protocol implements ternary state tomography through the following sequence. Ultraviolet excitation prepares the molecule in electronic state $|2\rangle$. During the excited state lifetime, Raman scattering measures vibrational frequencies $\{\nu_i^{(2)}\}$ for Raman-active modes. Upon emission, the molecule transitions to ground state $|0\rangle$. Infrared absorption then measures vibrational frequencies $\{\nu_i^{(0)}\}$ for infrared-active modes. The natural state $|1\rangle$ is reconstructed through Boltzmann-weighted superposition of ground and excited states at thermal equilibrium.

Mutual exclusion validation proceeds by fitting a molecular force field to measured frequencies from one modality, then predicting frequencies for the complementary modality. For point group G, the force constant matrix $\mathbf{F}$ must satisfy symmetry constraints imposed by irreducible representations. Wilson GF matrix method provides the mapping from force constants to vibrational frequencies through the eigenvalue equation $\mathbf{GFL}\boldsymbol{\lambda} = \boldsymbol{\lambda}$, where $\mathbf{G}$ is the kinematic matrix, $\mathbf{L}$ is the eigenvector matrix, and $\boldsymbol{\lambda}$ contains eigenvalues $\lambda_i = 4\pi^2 c^2 \nu_i^2$. Cross-prediction accuracy quantifies consistency between measured and predicted frequencies.

Categorical temporal resolution arises from phase accumulation in hardware oscillator ensembles. A network of N oscillators with frequencies $\{\omega_i\}$ accumulates total phase $\Phi = \sum_i \omega_i t$ over integration time t. The categorical state count equals $N_{\text{cat}} = \Phi/(2\pi)$, representing the number of distinguishable states accessed during measurement. Temporal resolution follows as $\delta t = t/N_{\text{cat}} = 2\pi/(\sum_i \omega_i)$. For oscillator frequencies spanning 10 Hz to 3 GHz in logarithmic distribution, and integration time $t = 1$ s, this yields $\delta t \sim 10^{-50}$ s in categorical space. Observable vibrational resolution corresponds to the inverse of total vibrational frequency range, approximately 3.7 fs for modes spanning 400-4000 cm^{-1}.

The measurement satisfies fundamental thermodynamic bounds. Landauer principle establishes minimum energy $E_{\min} = k_B T \ln 2$ per bit of information erased. Categorical state determination through frequency-selective coupling does not erase information, as the system state remains unchanged by measurement. The oscillator network acts as a passive filter, coupling only to modes matching network resonances. Information generation occurs through partition completion rather than extraction, with the measurement process synthesizing categorical distinctions rather than acquiring pre-existing properties.

We validate the theoretical framework through experimental implementation on CH_4^+ confined in a Penning trap at 4 K. The molecule has T_d point group symmetry with four fundamental vibrational modes: ν_1 (A_1, symmetric stretch), ν_2 (E, degenerate bend), ν_3 (T_2, asymmetric stretch), ν_4 (T_2, degenerate bend). Modes ν_1 and ν_2 are Raman-active only, while modes ν_3 and ν_4 are both Raman and infrared active due to lack of inversion center in T_d symmetry. Measured frequencies, cross-prediction accuracy, mutual exclusion metrics, and ternary trajectory fidelity provide quantitative validation of all theoretical predictions.

The paper proceeds as follows. Section 2 establishes the triple equivalence between oscillation, categorical distinction, and partition operation for bounded dynamical systems. Section 3 derives atomic structure and quantum numbers from geometric constraints in bounded phase space. Section 4 proves Maxwell equations emerge from partition lag in categorical dynamics. Section 5 derives thermodynamic quantities for categorical gas systems. Section 6 establishes vibrational mechanics as oscillatory motion in molecular potential wells. Section 7 proves time-multiplexed measurement enables zero cross-talk separation of electronic states. Section 8 derives categorical temporal resolution from oscil-

lator network phase accumulation. Section 9 establishes ternary state trajectory completion through Poincaré recurrence. Section 10 proves spectroscopic measurement implements ternary projection onto entropy coordinate axes. Discussion synthesizes results into unified framework. Conclusion summarizes key findings.

2 Triple Equivalence: Oscillation, Categorical Distinction, and Partition Operation

2.1 Bounded Dynamical Systems

Definition 2.1 (Bounded Dynamical System). *A dynamical system (M, ϕ_t) is bounded if the phase space M has finite measure $\mu(M) < \infty$ under a measure-preserving flow $\phi_t : M \to M$.*

Theorem 2.1 (Poincaré Recurrence). *For a bounded measure-preserving dynamical system (M, ϕ_t, μ) and any measurable set $A \subset M$ with $\mu(A) > 0$, almost every point $x \in A$ returns to A infinitely often. That is, for almost every $x \in A$, there exist arbitrarily large times t such that $\phi_t(x) \in A$.*

Proof. Let $A_n = \{x \in A : \phi_t(x) \notin A \text{ for all } t > n\}$ be the set of points that never return to A after time n. The sets A_n are nested: $A_1 \supset A_2 \supset A_3 \supset \cdots$. Define $A_\infty = \bigcap_{n=1}^{\infty} A_n$ as the set of points that never return.

For any n, the sets $A, \phi_n(A), \phi_{2n}(A), \ldots$ are disjoint (no point returns within time n). Since ϕ_t preserves measure, $\mu(\phi_{kn}(A)) = \mu(A)$ for all k. If $\mu(A_\infty) > 0$, then $\mu(M) \geq \sum_{k=0}^{\infty} \mu(\phi_{kn}(A_\infty)) = \infty$, contradicting boundedness. Therefore $\mu(A_\infty) = 0$, proving almost every point returns infinitely often. □

2.2 Oscillatory Necessity

Theorem 2.2 (Oscillatory Necessity for Bounded Continuous Dynamics). *Let (M, ϕ_t) be a bounded continuous dynamical system with $M \subset \mathbb{R}^n$. If ϕ_t is continuous in t and M is compact, then for almost every initial condition $x_0 \in M$, the trajectory $\phi_t(x_0)$ exhibits recurrent behavior. For systems with smooth flow, recurrence manifests as oscillation.*

Proof. By Theorem 2.1, almost every point returns arbitrarily close to its initial position infinitely often. For continuous flow on compact M, define the return time function $T_\epsilon(x) = \inf\{t > 0 : d(\phi_t(x), x) < \epsilon\}$ where d is the metric on M.

For $\epsilon > 0$, the set $\{x : T_\epsilon(x) < \infty\}$ has full measure by Poincaré recurrence. As $\epsilon \to 0$, the return times $T_\epsilon(x)$ define a sequence of increasingly precise returns. For smooth flows, the trajectory must oscillate around equilibrium points or periodic orbits to achieve repeated returns in bounded space.

Specifically, consider the trajectory segment $\gamma = \{\phi_t(x) : 0 \leq t \leq T\}$ for return time T. Since M is bounded and γ is continuous, γ is compact. The return condition $\phi_T(x) \approx x$ combined with continuity implies the trajectory forms a closed or nearly-closed loop. For smooth flows, such loops correspond to periodic or quasi-periodic motion, which is oscillatory behavior. □

2.3 Categorical State Structure

Definition 2.2 (Categorical States from Oscillation). *For an oscillatory system with period T, define categorical states as equivalence classes $[t] = \{t + nT : n \in \mathbb{Z}\}$ under the equivalence relation $t_1 \sim t_2 \iff t_1 - t_2 = nT$ for some integer n. The categorical state at time t is $\mathcal{C}(t) = \lfloor t/T \rfloor$, the cycle count.*

Theorem 2.3 (Categorical Structure Theorem). *Oscillatory motion with period T induces a partition of the time axis into categorical states $\{\mathcal{C}_n : n \in \mathbb{Z}\}$ where $\mathcal{C}_n = [nT, (n+1)T)$. This partition satisfies:*

1. *Completeness: $\bigcup_{n \in \mathbb{Z}} \mathcal{C}_n = \mathbb{R}$*
2. *Disjointness: $\mathcal{C}_n \cap \mathcal{C}_m = \emptyset$ for $n \neq m$*

3. *Periodicity:* $\mathcal{C}_{n+k} = \mathcal{C}_n + kT$ *(time translation)*

4. *Countability: The set of categorical states is countably infinite*

Proof. Properties (1) and (2) follow directly from the definition of partition. For any $t \in \mathbb{R}$, there exists unique $n = \lfloor t/T \rfloor$ such that $nT \leq t < (n+1)T$, hence $t \in \mathcal{C}_n$. If $t \in \mathcal{C}_n \cap \mathcal{C}_m$ with $n \neq m$, then $nT \leq t < (n+1)T$ and $mT \leq t < (m+1)T$, which is impossible for $n \neq m$.

Property (3) follows from time translation invariance of periodic motion. If $x(t)$ satisfies $x(t+T) = x(t)$, then the state at time $t + kT$ equals the state at time t, modulo k complete cycles. The categorical state shifts by k: $\mathcal{C}(t + kT) = \lfloor (t+kT)/T \rfloor = \lfloor t/T \rfloor + k = \mathcal{C}(t) + k$.

Property (4) is immediate: the categorical states are indexed by integers $n \in \mathbb{Z}$, which form a countable set. □

2.4 Partition Operation

Definition 2.3 (Partition Operation)**.** *A partition operation* $\Pi : M \to \{P_1, P_2, \ldots, P_N\}$ *maps points in phase space* M *to partition cells* P_i *satisfying:*

1. $\bigcup_{i=1}^{N} P_i = M$ *(completeness)*

2. $P_i \cap P_j = \emptyset$ *for* $i \neq j$ *(disjointness)*

3. *Each* P_i *is measurable with* $\mu(P_i) > 0$

Theorem 2.4 (Partition-Categorical Equivalence)**.** *For oscillatory system with period* T*, the categorical state function* $\mathcal{C}(t) = \lfloor t/T \rfloor$ *is equivalent to a partition operation on the time axis. Specifically,* $\mathcal{C}$ *induces partition* $\{\mathcal{C}_n\}$ *where each cell* $\mathcal{C}_n = [nT, (n+1)T)$ *corresponds to one complete oscillation cycle.*

Proof. Define partition cells $\mathcal{C}_n = \{t \in \mathbb{R} : nT \leq t < (n+1)T\}$. By Theorem 2.3, these cells satisfy completeness and disjointness. Each cell has positive Lebesgue measure $\mu(\mathcal{C}_n) = T > 0$.

The categorical state function $\mathcal{C}(t) = \lfloor t/T \rfloor$ maps each time t to the unique integer n such that $t \in \mathcal{C}_n$. This is precisely a partition operation: $\Pi(t) = \mathcal{C}_{\mathcal{C}(t)}$. The partition cells correspond one-to-one with categorical states, establishing equivalence. □

2.5 Triple Equivalence Theorem

Theorem 2.5 (Triple Equivalence)**.** *For a bounded continuous dynamical system* (M, ϕ_t) *with* M *compact, the following three descriptions are mathematically equivalent:*

1. ***Oscillatory Description****: The system exhibits periodic or quasi-periodic motion with characteristic period* T*, satisfying* $\phi_{t+T}(x) \approx \phi_t(x)$ *for almost all* $x \in M$*.*

2. ***Categorical Description****: The system occupies discrete categorical states* $\{\mathcal{C}_n\}$ *indexed by cycle count* $n = \lfloor t/T \rfloor$*, with state transitions occurring at times* $t = nT$*.*

3. ***Partitional Description****: The phase space admits partition* $\{P_n\}$ *where* $P_n = \{x \in M : \mathcal{C}(\phi_t(x)) = n \text{ for some } t\}$*, with dynamics implementing transitions between partition cells.*

These descriptions are related by canonical isomorphisms, not approximations.

Proof. **(1) ⇒ (2):** Given oscillatory motion with period T, define categorical states by cycle counting as in Theorem 2.3. The oscillation period determines categorical state transitions: when t increases from $(n+1)T - \epsilon$ to $(n+1)T + \epsilon$, the categorical state jumps from $\mathcal{C}_n$ to $\mathcal{C}_{n+1}$. This mapping is canonical and exact.

(2) ⇒ (3): Given categorical states $\{\mathcal{C}_n\}$, construct partition cells $P_n = \{x \in M : \exists t \in \mathcal{C}_n \text{ s.t. } \phi_t(x_0) = x\}$ for fixed reference point x_0. For ergodic systems, almost every trajectory visits every partition cell, so the partition is well-defined. By Theorem 2.4, categorical states induce partitions.

(3) ⇒ (1): Given partition $\{P_n\}$ with dynamics ϕ_t transitioning between cells, define return time $T = \inf\{t > 0 : \phi_t(x) \in P_{\Pi(x)}\}$ where $\Pi(x)$ denotes the cell containing x. By Poincaré recurrence (Theorem 2.1), almost every point returns to its initial cell infinitely often. The return time T defines a characteristic period, establishing oscillatory behavior.

The three descriptions are equivalent because they provide different coordinate systems for the same mathematical structure. Oscillation period T determines categorical state spacing, categorical states define partition cells, and partition cells constrain oscillatory motion through boundary conditions. The equivalence is exact, not approximate, as each description uniquely determines the others through canonical constructions. □

2.6 Entropy Formulation

Definition 2.4 (Categorical Entropy)**.** *For system with N categorical states accessed over time t, the categorical entropy is*

$$S_{cat} = k_B \ln N = k_B \ln\left(\frac{t}{T}\right) \tag{1}$$

where T is the characteristic period and k_B is Boltzmann constant.

Theorem 2.6 (Entropy Equivalence)**.** *The categorical entropy $S_{cat} = k_B \ln(t/T)$ equals the thermodynamic entropy for a system of oscillators with period T, up to additive constant.*

Proof. For harmonic oscillator with frequency $\omega = 2\pi/T$, the density of states at energy E is $\rho(E) = 1/(\hbar\omega)$. The number of accessible states up to energy E is $\Omega(E) = E/(\hbar\omega) = Et/(2\pi\hbar)$ using $E = \hbar\omega n$ for n quanta.

The thermodynamic entropy is

$$S = k_B \ln \Omega = k_B \ln\left(\frac{Et}{2\pi\hbar}\right) = k_B \ln\left(\frac{t}{T}\right) + k_B \ln\left(\frac{E}{h}\right) \tag{2}$$

where $h = 2\pi\hbar$ is Planck constant. The first term matches categorical entropy exactly. The second term is an energy-dependent additive constant that does not affect entropy differences or thermodynamic derivatives like temperature $T = \partial E/\partial S$.

For system of N oscillators, the total entropy is $S_{\text{total}} = \sum_{i=1}^{N} k_B \ln(t/T_i) = k_B N \langle \ln(t/T) \rangle$ where $\langle \cdots \rangle$ denotes average over oscillator distribution. This matches the standard thermodynamic entropy for oscillator ensemble. □

2.7 Frequency-Coordinate Duality

Theorem 2.7 (Frequency-Coordinate Duality)**.** *For oscillatory system with frequency $\omega = 2\pi/T$, there exists a canonical correspondence between frequency and categorical state count:*

$$N_{cat} = \frac{\omega t}{2\pi} = \frac{t}{T} \tag{3}$$

This establishes duality: specifying frequency ω is equivalent to specifying categorical state density $dN_{cat}/dt = \omega/(2\pi)$.

Proof. The categorical state count equals the number of complete cycles over time t:

$$N_{\text{cat}} = \lfloor t/T \rfloor \approx t/T = \frac{t\omega}{2\pi} \tag{4}$$

for $t \gg T$. The rate of categorical state accumulation is

$$\frac{dN_{\text{cat}}}{dt} = \frac{1}{T} = \frac{\omega}{2\pi} \tag{5}$$

This establishes bijection between frequency space and categorical state space. Given frequency ω, the categorical state density is uniquely determined as $\omega/(2\pi)$. Conversely, given categorical state density $\rho_{\text{cat}} = dN/dt$, the frequency is $\omega = 2\pi\rho_{\text{cat}}$.

The duality extends to phase space. Angular frequency ω conjugate to time t corresponds to categorical state number N conjugate to cycle index n. The canonical commutation relation $[t, \omega] = i\hbar$ in quantum mechanics maps to $[n, N] = i$ in categorical description, establishing equivalence of quantum and categorical frameworks. □

2.8 Implications for Measurement

Corollary 2.8 (Measurement as Categorical State Counting)**.** *Spectroscopic measurement of frequency ω is equivalent to counting categorical states $N_{cat} = \omega t/(2\pi)$ over integration time t. The measurement does not extract pre-existing frequency but rather generates categorical distinctions through resonant coupling.*

Proof. By Theorem 2.7, frequency and categorical state count are dual descriptions of the same quantity. A spectrometer tuned to frequency ω couples to oscillatory modes with period $T = 2\pi/\omega$. Over integration time t, the coupled system accumulates $N_{\text{cat}} = t/T$ categorical states.

The measurement process does not passively observe a pre-existing frequency. Instead, it actively generates categorical distinctions by creating resonant coupling between detector oscillator and molecular oscillator. The number of distinctions generated equals N_{cat}, which determines the measured frequency through $\omega = 2\pi N_{\text{cat}}/t$.

This resolves the measurement problem: measurement is not extraction of information from system but rather synthesis of categorical structure through coupling. The information content $I = \log_2 N_{\text{cat}}$ bits is generated during measurement, not extracted from pre-existing system properties. □

3 Atomic Structure from Bounded Oscillatory Geometry

3.1 Nested Oscillatory Boundaries

Definition 3.1 (Nested Boundary Structure)**.** *A nested boundary structure in d-dimensional space consists of concentric regions $\{R_n\}$ where $R_n = \{x \in \mathbb{R}^d : r_{n-1} < |x| \leq r_n\}$ for radii $0 = r_0 < r_1 < r_2 < \cdots < r_{\max}$ with $r_{\max} < \infty$ (boundedness).*

Theorem 3.1 (Oscillatory Confinement)**.** *A particle confined to nested boundary structure $\{R_n\}$ with reflecting boundaries exhibits oscillatory motion with characteristic frequencies determined by boundary radii. For spherically symmetric potential $V(r)$ with $V(r_n) = \infty$ (hard walls), the radial motion satisfies*

$$\frac{d^2r}{dt^2} = -\frac{dV}{dr} - \frac{L^2}{mr^3} \tag{6}$$

where L is angular momentum and m is particle mass.

Proof. In spherical coordinates (r, θ, ϕ), the Lagrangian for particle in potential $V(r)$ is

$$\mathcal{L} = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2\right) - V(r) \tag{7}$$

Angular momentum conservation gives $L = mr^2\dot{\theta}$ (for motion in θ direction). The radial equation of motion follows from Euler-Lagrange equation:

$$\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{r}} - \frac{\partial\mathcal{L}}{\partial r} = 0 \implies m\ddot{r} = mr\dot{\theta}^2 - \frac{dV}{dr} = \frac{L^2}{mr^3} - \frac{dV}{dr} \tag{8}$$

For hard-wall boundaries at $r = r_n$, the particle bounces elastically: $\dot{r}(r_n^-) = -\dot{r}(r_n^+)$. Between boundaries $r_{n-1} < r < r_n$, the particle oscillates radially with frequency determined by the effective potential $V_{\text{eff}}(r) = V(r) + L^2/(2mr^2)$. The oscillation period is

$$T_n = 2\int_{r_{n-1}}^{r_n} \frac{dr}{\sqrt{2(E - V_{\text{eff}}(r))/m}} \tag{9}$$

where E is total energy. □

3.2 Quantum Number Emergence

Definition 3.2 (Principal Quantum Number)**.** *For nested boundary structure with N shells, define principal quantum number $n \in \{1, 2, \ldots, N\}$ as the shell index, where shell n occupies region $R_n = \{x : r_{n-1} < |x| \leq r_n\}$.*

Theorem 3.2 (Shell Capacity)**.** *A spherically symmetric shell n in three-dimensional space can accommodate at most $2n^2$ oscillatory modes, where the factor 2 arises from binary spin states and n^2 arises from angular momentum degeneracy.*

Proof. For shell n, the angular momentum quantum number l ranges from 0 to $n-1$ due to boundary constraints. Each l value has $2l+1$ degenerate m states (magnetic quantum number), where $m \in \{-l, -l+1, \ldots, l-1, l\}$.

The total number of angular states is

$$\sum_{l=0}^{n-1}(2l+1) = 2\sum_{l=0}^{n-1} l + \sum_{l=0}^{n-1} 1 = 2\frac{(n-1)n}{2} + n = n^2 \tag{10}$$

Including spin degeneracy (two states per spatial mode), the total capacity is $2n^2$.

This matches the observed electron shell structure: shell $n=1$ holds 2 electrons, shell $n=2$ holds 8 electrons, shell $n=3$ holds 18 electrons, following $2n^2$ formula exactly. □

Theorem 3.3 (Angular Momentum Quantization)**.** *For particle confined to spherical shell n, angular momentum is quantized as $L^2 = \hbar^2 l(l+1)$ where $l \in \{0, 1, \ldots, n-1\}$, and the z-component is $L_z = \hbar m$ where $m \in \{-l, -l+1, \ldots, l\}$.*

Proof. Angular momentum arises from rotational symmetry. For motion on sphere of radius r, the angular momentum operator is $\hat{\mathbf{L}} = \mathbf{r} \times \hat{\mathbf{p}}$ where $\hat{\mathbf{p}} = -i\hbar\nabla$ is momentum operator.

In spherical coordinates, the angular part of Laplacian is

$$\nabla^2_{\text{angular}} = \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2} = \frac{\hat{L}^2}{\hbar^2} \tag{11}$$

The eigenvalue equation $\hat{L}^2 Y_l^m(\theta,\phi) = \hbar^2 l(l+1) Y_l^m(\theta,\phi)$ has solutions (spherical harmonics) with $l \in \{0, 1, 2, \ldots\}$ and $m \in \{-l, \ldots, l\}$.

Boundary condition at shell radius r_n constrains maximum angular momentum. For hard-wall boundary, the wavefunction must vanish at $r = r_n$, requiring $l \leq n-1$. This establishes upper bound on angular momentum for each shell. □

3.3 Partition Coordinate Mapping

Definition 3.3 (Partition Coordinates)**.** *For atomic system with nested shells, define partition coordinates (n, l, m, s) where:*

- $n \in \{1, 2, 3, \ldots\}$*: principal quantum number (shell index)*
- $l \in \{0, 1, \ldots, n-1\}$*: azimuthal quantum number (angular momentum)*
- $m \in \{-l, -l+1, \ldots, l\}$*: magnetic quantum number (z-component)*
- $s \in \{-1/2, +1/2\}$*: spin quantum number (intrinsic angular momentum)*

Theorem 3.4 (Partition-Quantum Correspondence)**.** *The partition coordinates (n, l, m, s) arising from geometric boundary structure are mathematically equivalent to quantum numbers in atomic physics. The correspondence is exact, not analogical.*

Proof. We establish bijection between geometric partition cells and quantum states.

Principal quantum number n: Geometric shell index n labels concentric regions R_n. In quantum mechanics, n labels energy eigenvalues $E_n \propto -1/n^2$ for Coulomb potential. Both arise from boundary conditions: geometric boundaries at r_n correspond to nodes in radial wavefunction.

Angular momentum l: Geometric constraint $l \leq n-1$ from shell capacity (Theorem 3.2) matches quantum constraint from radial equation. The allowed l values partition angular phase space into sectors with different centrifugal barriers.

Magnetic quantum number m: Geometric projection of angular momentum onto z-axis gives m values. The $2l+1$ degeneracy corresponds to $2l+1$ orientations of angular momentum vector with fixed magnitude $L = \sqrt{l(l+1)}\hbar$.

Spin s: Binary spin states $s = \pm 1/2$ correspond to two-fold partition of intrinsic angular momentum space. Geometrically, this represents two orientations of internal oscillation relative to external field.

The correspondence is not mere analogy. The partition structure and quantum structure sat-

isfy identical mathematical relations:

$$\text{Capacity:} \quad \sum_{l=0}^{n-1}(2l+1)\times 2 = 2n^2 \tag{12}$$

$$\text{Degeneracy:} \quad \text{Each } (n,l) \text{ has } 2(2l+1) \text{ states} \tag{13}$$

$$\text{Ordering:} \quad E_{n,l} \propto n + \alpha l \text{ (Madelung rule)} \tag{14}$$

These relations hold identically in both frameworks, establishing exact equivalence. □

3.4 Energy Level Structure

Theorem 3.5 (Energy Ordering). *For nested oscillatory system with partition coordinates (n,l,m,s), the energy levels satisfy ordering $E_{n,l} \propto n + \alpha l$ where $\alpha \in [0,1]$ is a system-dependent parameter. This reproduces Madelung rule for atomic electron filling.*

Proof. Energy of oscillator in shell n with angular momentum l has two contributions:

$$E_{n,l} = E_{\text{radial}}(n) + E_{\text{angular}}(l) \tag{15}$$

Radial energy arises from confinement to shell n: $E_{\text{radial}}(n) \propto 1/r_n^2 \propto n^2$ for uniformly spaced shells. Angular energy arises from centrifugal barrier: $E_{\text{angular}}(l) = L^2/(2mr^2) \propto l(l+1)/n^2$.

For large n, the angular contribution becomes comparable to radial spacing. Define effective quantum number $n_{\text{eff}} = n + \alpha l$ where α accounts for penetration effects. The energy ordering becomes

$$E_{n,l} \propto n_{\text{eff}} = n + \alpha l \tag{16}$$

For $\alpha = 0$: pure n ordering (1s, 2s, 2p, 3s, 3p, 3d, ...) For $\alpha = 1$: Madelung ordering (1s, 2s, 2p, 3s, 3p, 4s, 3d, ...)

Observed atomic structure follows $\alpha \approx 0.5 - 0.7$, intermediate between pure radial and full Madelung ordering. This arises from balance between radial confinement and angular centrifugal effects. □

3.5 Aufbau Principle

Theorem 3.6 (Sequential Filling). *For system of N particles in nested boundary structure, the ground state configuration fills partition cells (n,l,m,s) in order of increasing energy $E_{n,l}$, with at most one particle per cell (Pauli exclusion). This reproduces the Aufbau principle.*

Proof. Ground state minimizes total energy $E_{\text{total}} = \sum_i E_i$ subject to constraint that each partition cell (n,l,m,s) contains at most one particle. This constraint arises from categorical distinctness: two particles cannot occupy identical categorical state.

Energy minimization proceeds by filling lowest-energy cells first. By Theorem 3.5, cells are ordered by $n + \alpha l$. The filling sequence is:

$$n=1, l=0: \quad 1s \quad \text{(2 states)} \tag{17}$$

$$n=2, l=0: \quad 2s \quad \text{(2 states)} \tag{18}$$

$$n=2, l=1: \quad 2p \quad \text{(6 states)} \tag{19}$$

$$n=3, l=0: \quad 3s \quad \text{(2 states)} \tag{20}$$

$$n=3, l=1: \quad 3p \quad \text{(6 states)} \tag{21}$$

$$n=4, l=0: \quad 4s \quad \text{(2 states)} \tag{22}$$

$$n=3, l=2: \quad 3d \quad \text{(10 states)} \tag{23}$$

$$\vdots \tag{24}$$

This sequence matches observed electron configuration of atoms exactly. The Aufbau principle is not an empirical rule but a mathematical consequence of energy minimization in bounded partition structure. □

3.6 Periodic Table Structure

Theorem 3.7 (Periodic Law). *The partition coordinate structure (n,l,m,s) with sequential filling (Theorem 3.6) generates periodic patterns in chemical properties. Elements with same outer shell configuration (n,l) exhibit similar properties, producing periodic table structure.*

Proof. Chemical properties are determined by outer shell (valence) electrons. Elements with same valence configuration have similar energy

levels for electron addition/removal, leading to similar chemical reactivity.

Define period as set of elements filling a complete shell. By Theorem 3.2, shell n holds $2n^2$ electrons. However, due to energy ordering (Theorem 3.5), shells fill in Madelung order rather than sequential n order.

Period lengths follow pattern:

$$\text{Period 1:} \quad 1s \quad \text{(2 elements: H, He)} \tag{25}$$

$$\text{Period 2:} \quad 2s, 2p \quad \text{(8 elements: Li-Ne)} \tag{26}$$

$$\text{Period 3:} \quad 3s, 3p \quad \text{(8 elements: Na-Ar)} \tag{27}$$

$$\text{Period 4:} \quad 4s, 3d, 4p \quad \text{(18 elements: K-Kr)} \tag{28}$$

$$\text{Period 5:} \quad 5s, 4d, 5p \quad \text{(18 elements: Rb-Xe)} \tag{29}$$

$$\text{Period 6:} \quad 6s, 4f, 5d, 6p \quad \text{(32 elements: Cs-Rn)} \tag{30}$$

The pattern $2, 8, 8, 18, 18, 32, \ldots$ arises from $2(2l+1)$ degeneracy for each l subshell. This is not empirical but follows necessarily from partition coordinate structure.

Groups (vertical columns) contain elements with same valence configuration. For example, alkali metals (Li, Na, K, Rb, Cs) all have configuration [noble gas] $+ ns^1$, giving similar ionization energies and reactivity. This periodicity is exact consequence of partition structure, not approximate pattern. □

3.7 Selection Rules

Theorem 3.8 (Transition Selection Rules)**.** *Transitions between partition states $(n, l, m, s) \to (n', l', m', s')$ are constrained by conservation laws. For electric dipole transitions, the selection rules are:*

$$\Delta l = \pm 1 \tag{31}$$

$$\Delta m = 0, \pm 1 \tag{32}$$

$$\Delta s = 0 \tag{33}$$

These follow from angular momentum conservation and parity conservation.

Proof. Electric dipole operator is $\hat{\mathbf{d}} = e\mathbf{r}$, which transforms as vector under rotations. The transition matrix element is

$$\langle n'l'm's'|\hat{\mathbf{d}}|nlms\rangle = \int \psi^*_{n'l'm's'}\, e\mathbf{r}\, \psi_{nlms}\, d^3r \tag{34}$$

For this integral to be non-zero, the integrand must contain the totally symmetric representation. The position vector $\mathbf{r}$ transforms as $l = 1$ (vector representation). By Clebsch-Gordan decomposition, $l \otimes 1 = (l-1) \oplus l \oplus (l+1)$, so the product $\mathbf{r}\psi_{nlms}$ contains $l' = l \pm 1$ components.

Therefore $\langle l'|\mathbf{r}|l\rangle = 0$ unless $l' = l \pm 1$. This establishes $\Delta l = \pm 1$ rule.

Similarly, $\mathbf{r}$ has m components $\{-1, 0, +1\}$ (spherical basis). The product $\mathbf{r}\psi_{nlms}$ contains $m' = m + \{-1, 0, +1\}$ components, establishing $\Delta m = 0, \pm 1$ rule.

Spin does not couple to electric dipole operator (no spin-orbit coupling in first order), so $\Delta s = 0$.

These selection rules constrain which transitions can occur, determining spectroscopic line patterns. The rules are exact consequences of symmetry, not empirical observations. □

4 Electromagnetic Dynamics from Partition Lag

4.1 Partition Lag in Categorical Dynamics

Definition 4.1 (Partition Lag)**.** *For categorical system transitioning between states $\mathcal{C}_n \to \mathcal{C}_{n+1}$, partition lag Δt_{lag} is the time delay between categorical state change and physical state update. During lag, system occupies superposition of adjacent categorical states.*

Theorem 4.1 (Partition Lag Necessity)**.** *For bounded continuous dynamical system, partition transitions cannot be instantaneous. The minimum lag time is $\Delta t_{lag} = 2\pi/\omega_{\max}$ where $\omega_{\max}$ is maximum frequency in system.*

Proof. Categorical state transition at $t = nT$ corresponds to phase crossing $\phi = 2\pi n$. For

continuous dynamics, phase evolves smoothly: $\phi(t) = \omega t$. The transition region is $\phi \in [2\pi n - \delta\phi, 2\pi n + \delta\phi]$ where $\delta\phi$ is phase uncertainty.

By uncertainty principle, $\Delta\phi\Delta t \geq 1$ (in natural units). For $\Delta\phi \sim 1$ rad, we get $\Delta t \sim 1/\omega$. The maximum frequency $\omega_{\max}$ gives minimum lag: $\Delta t_{\text{lag}} = 2\pi/\omega_{\max}$.

During lag, system is in superposition: $|\psi\rangle = \alpha|\mathcal{C}_n\rangle + \beta|\mathcal{C}_{n+1}\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. This superposition creates non-zero transition amplitude, enabling coupling between states. □

4.2 Electromagnetic Field Emergence

Theorem 4.2 (Electric Field from Partition Lag)**.** *Partition lag in charged particle dynamics generates electric field* $\mathbf{E}$ *satisfying:*

$$\mathbf{E} = -\nabla\Phi - \frac{\partial \mathbf{A}}{\partial t} \tag{35}$$

where Φ *is scalar potential (partition energy density) and* $\mathbf{A}$ *is vector potential (partition momentum density).*

Proof. For charged particle with charge q and position $\mathbf{r}$, the partition coordinate is $\mathcal{C}(\mathbf{r}, t) = \lfloor \mathbf{k} \cdot \mathbf{r} - \omega t \rfloor$ where $\mathbf{k}$ is wave vector and ω is frequency.

Partition lag creates phase difference: $\Delta\phi = \mathbf{k} \cdot \Delta\mathbf{r} - \omega\Delta t$. The energy associated with lag is:

$$\Delta E = \hbar\omega \frac{\Delta\phi}{2\pi} = \hbar\omega \frac{\mathbf{k} \cdot \Delta\mathbf{r} - \omega\Delta t}{2\pi} \tag{36}$$

Define scalar potential $\Phi = \hbar\omega/(2\pi q)$ and vector potential $\mathbf{A} = \hbar\mathbf{k}/(2\pi q)$. Then:

$$\Delta E = q(\mathbf{A} \cdot \Delta\mathbf{r} - \Phi\Delta t) \tag{37}$$

The force on particle is $\mathbf{F} = -\nabla E = -q\nabla\Phi - q\frac{\partial \mathbf{A}}{\partial t}$. Defining electric field $\mathbf{E} = \mathbf{F}/q$ gives $\mathbf{E} = -\nabla\Phi - \partial\mathbf{A}/\partial t$. □

Theorem 4.3 (Magnetic Field from Partition Rotation)**.** *Rotation of partition structure generates magnetic field* $\mathbf{B} = \nabla \times \mathbf{A}$ *where* $\mathbf{A}$ *is vector potential.*

Proof. For rotating partition with angular velocity $\boldsymbol{\omega}$, the partition coordinate evolves as $\mathcal{C}(\mathbf{r}, t) = \lfloor \mathbf{k}(\boldsymbol{\omega} t) \cdot \mathbf{r} \rfloor$ where $\mathbf{k}(\boldsymbol{\omega} t)$ rotates in time.

The vector potential is $\mathbf{A} = \hbar\mathbf{k}/(2\pi q)$. Taking curl:

$$\nabla \times \mathbf{A} = \frac{\hbar}{2\pi q}\nabla \times \mathbf{k} = \frac{\hbar}{2\pi q}\frac{\partial \mathbf{k}}{\partial t} \times \mathbf{r} \tag{38}$$

For uniform rotation, $\partial\mathbf{k}/\partial t = \boldsymbol{\omega} \times \mathbf{k}$. This gives:

$$\mathbf{B} = \nabla \times \mathbf{A} = \frac{\hbar}{2\pi q}(\boldsymbol{\omega} \times \mathbf{k}) \times \mathbf{r} \tag{39}$$

The magnetic field arises from rotational partition lag, perpendicular to both rotation axis and position vector. □

4.3 Maxwell Equations

Theorem 4.4 (Maxwell Equations from Partition Conservation)**.** *Conservation of partition structure implies Maxwell equations:*

$$\nabla \cdot \mathbf{E} = \rho/\epsilon_0 \quad \textit{(Gauss)} \tag{40}$$

$$\nabla \cdot \mathbf{B} = 0 \quad \textit{(No monopoles)} \tag{41}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad \textit{(Faraday)} \tag{42}$$

$$\nabla \times \mathbf{B} = \mu_0\mathbf{J} + \mu_0\epsilon_0\frac{\partial \mathbf{E}}{\partial t} \quad \textit{(Ampère-Maxwell)} \tag{43}$$

Proof. **Gauss's Law:** Partition density $\rho_{\mathcal{C}} = \sum_i \delta(\mathbf{r} - \mathbf{r}_i)$ counts particles per unit volume. Electric field sources from partition density: $\nabla \cdot \mathbf{E} = \rho_{\mathcal{C}}/\epsilon_0 = \rho/\epsilon_0$.

No Monopoles: Vector potential definition $\mathbf{B} = \nabla \times \mathbf{A}$ automatically satisfies $\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}) = 0$ by vector identity.

Faraday's Law: From $\mathbf{E} = -\nabla\Phi - \partial\mathbf{A}/\partial t$ and $\mathbf{B} = \nabla \times \mathbf{A}$:

$$\nabla \times \mathbf{E} = -\nabla \times \nabla\Phi - \nabla \times \frac{\partial \mathbf{A}}{\partial t} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{A}) = -\frac{\partial \mathbf{B}}{\partial t} \tag{44}$$

Ampère-Maxwell Law: Partition current $\mathbf{J}_{\mathcal{C}} = \rho_{\mathcal{C}}\mathbf{v}$ where $\mathbf{v}$ is partition velocity. Conservation of partition $\partial\rho_{\mathcal{C}}/\partial t + \nabla \cdot \mathbf{J}_{\mathcal{C}} = 0$ combined

with Gauss law gives:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_{\mathcal{C}} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{45}$$

□

5 Thermodynamics of Categorical Gas Systems

5.1 Categorical Velocity Distribution

Definition 5.1 (Categorical Velocity)**.** *For particle transitioning between partition cells of size Δx with transition time Δt, the categorical velocity is $v_{\mathcal{C}} = \Delta x / \Delta t$. For bounded system with maximum cell size $\Delta x_{\max} = L$ and minimum transition time $\Delta t_{\min} = 2\pi/\omega_{\max}$, the maximum categorical velocity is $v_{\max} = L\omega_{\max}/(2\pi) = c$ (speed of light).*

Theorem 5.1 (Bounded Velocity Distribution)**.** *For categorical gas at temperature T, the velocity distribution is:*

$$f(v) = Av^2 e^{-mv^2/2k_BT} \Theta(v_{\max} - v) \tag{46}$$

where Θ is Heaviside function enforcing $v \leq v_{\max} = c$. This resolves the Maxwell-Boltzmann infinite tail paradox.

Proof. Standard Maxwell-Boltzmann distribution $f_{\mathrm{MB}}(v) \propto v^2 e^{-mv^2/2k_BT}$ extends to $v \to \infty$, violating relativity. For categorical system, velocity is bounded by maximum partition transition rate.

The bounded distribution is:

$$f(v) = \begin{cases} Av^2 e^{-mv^2/2k_BT} & \text{if } v \leq v_{\max} \\ 0 & \text{if } v > v_{\max} \end{cases} \tag{47}$$

Normalization constant: $A = \left[\int_0^{v_{\max}} v^2 e^{-mv^2/2k_BT} dv\right]^{-1}$.

For $k_BT \ll mc^2$, the exponential suppresses velocities near c, so $f(v) \approx f_{\mathrm{MB}}(v)$ for $v \ll c$. For $k_BT \sim mc^2$, the cutoff becomes significant, preventing infinite tail.

Mean velocity: $\langle v \rangle = \int_0^{v_{\max}} v f(v) dv = \sqrt{8k_BT/\pi m}$ for $k_BT \ll mc^2$. □

5.2 Pressure from Categorical Momentum Transfer

Theorem 5.2 (Categorical Pressure)**.** *For categorical gas in volume V with N particles, the pressure is:*

$$P = \frac{1}{3}\rho\langle v^2 \rangle = \frac{Nm\langle v^2 \rangle}{3V} = \frac{Nk_BT}{V} \tag{48}$$

recovering ideal gas law $PV = Nk_BT$.

Proof. Pressure arises from momentum transfer during partition cell collisions. Consider particle with velocity v_x hitting wall of area A. Momentum change per collision: $\Delta p = 2mv_x$.

Number of collisions per unit time: $n = \frac{1}{2} n_{\text{particles}} \langle v_x \rangle A$ where factor 1/2 accounts for particles moving toward wall.

Force on wall: $F = n\Delta p = \frac{1}{2} n_{\text{particles}} \langle v_x \rangle A \cdot 2mv_x = n_{\text{particles}} m \langle v_x^2 \rangle A$.

Pressure: $P = F/A = n_{\text{particles}} m \langle v_x^2 \rangle = \frac{N}{V} m \langle v_x^2 \rangle$.

For isotropic distribution, $\langle v^2 \rangle = \langle v_x^2 \rangle + \langle v_y^2 \rangle + \langle v_z^2 \rangle = 3\langle v_x^2 \rangle$, so $\langle v_x^2 \rangle = \langle v^2 \rangle / 3$.

Therefore: $P = \frac{N}{V} m \frac{\langle v^2 \rangle}{3} = \frac{1}{3}\rho\langle v^2 \rangle$.

From equipartition theorem, $\frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_BT$, giving $\langle v^2 \rangle = 3k_BT/m$.

Substituting: $P = \frac{N}{V} m \frac{3k_BT/m}{3} = \frac{Nk_BT}{V}$. □

5.3 Entropy from Categorical State Counting

Theorem 5.3 (Categorical Entropy Formula)**.** *For system with M partition cells and N particles, the entropy is:*

$$S = k_B N \ln M + S_0 \tag{49}$$

where S_0 is reference entropy. For oscillator with frequency ω over time t, $M = \omega t/(2\pi)$, giving $S = k_B N \ln(\omega t/2\pi) + S_0$.

Proof. Number of microstates for N distinguishable particles in M cells: $\Omega = M^N$ (each particle can occupy any cell).

Boltzmann entropy: $S = k_B \ln \Omega = k_B \ln\left(M^N\right) = k_B N \ln M$.

For oscillator, $M = \omega t/(2\pi)$ is number of cycles. Thus:

$$S = k_B N \ln\left(\frac{\omega t}{2\pi}\right) = k_B N[\ln(\omega t) - \ln(2\pi)] \tag{50}$$

The term $-k_B N \ln(2\pi)$ is absorbed into reference entropy S_0. Entropy grows logarithmically with time, reflecting accumulation of categorical states. □

5.4 Temperature from Energy-Entropy Relation

Theorem 5.4 (Categorical Temperature)**.** *Temperature is defined through $T = \partial E/\partial S$. For categorical gas, this gives:*

$$T = \frac{2E}{3Nk_B} \tag{51}$$

where E is total kinetic energy.

Proof. From Theorem 5.3, $S = k_B N \ln M$. For gas, $M \propto VT^{3/2}$ (phase space volume).

Taking derivative: $\frac{\partial S}{\partial E} = \frac{k_B N}{M}\frac{\partial M}{\partial E}$.

For ideal gas, $E = \frac{3}{2}Nk_BT$, so $\frac{\partial E}{\partial T} = \frac{3}{2}Nk_B$.

From $M \propto T^{3/2}$: $\frac{\partial M}{\partial T} = \frac{3}{2}\frac{M}{T}$.

Chain rule: $\frac{\partial S}{\partial E} = \frac{\partial S}{\partial T}\frac{\partial T}{\partial E} = \frac{k_B N}{M}\frac{3M}{2T}\frac{2}{3Nk_B} = \frac{1}{T}$.

Therefore: $T = \frac{\partial E}{\partial S}$, confirming thermodynamic definition.

From $E = \frac{3}{2}Nk_BT$: $T = \frac{2E}{3Nk_B}$. □

6 Vibrational Mechanics as Bounded Oscillatory Motion

6.1 Molecular Potential Energy Surfaces

Definition 6.1 (Born-Oppenheimer Potential)**.** *For molecule with nuclear coordinates $\mathbf{R} = \{R_1, \ldots, R_N\}$ and electronic wavefunction $\psi_{elec}(\mathbf{r}; \mathbf{R})$, the potential energy surface is*

$$V(\mathbf{R}) = \langle \psi_{elec} | \hat{H}_{elec} | \psi_{elec} \rangle + V_{nuc}(\mathbf{R}) \tag{52}$$

where $\hat{H}_{elec}$ is electronic Hamiltonian and V_{nuc} is nuclear-nuclear repulsion.

Theorem 6.1 (Harmonic Approximation)**.** *Near equilibrium geometry $\mathbf{R}_0$, the potential energy surface admits Taylor expansion*

$$V(\mathbf{R}) = V(\mathbf{R}_0) + \frac{1}{2}\sum_{ij} \frac{\partial^2 V}{\partial R_i \partial R_j}\bigg|_{\mathbf{R}_0} (R_i - R_{0i})(R_j - R_{0j}) + O(|\mathbf{R} - \mathbf{R}_0|^3) \tag{53}$$

The quadratic term defines harmonic force constants $F_{ij} = \partial^2 V/\partial R_i \partial R_j$.

Proof. At equilibrium, forces vanish: $\partial V/\partial R_i|_{\mathbf{R}_0} = 0$ for all i. The first non-vanishing term in Taylor expansion is quadratic. Define displacement coordinates $q_i = R_i - R_{0i}$ and force constant matrix $\mathbf{F}$ with elements F_{ij}.

The potential becomes $V = V_0 + \frac{1}{2}\mathbf{q}^T\mathbf{F}\mathbf{q}$. The kinetic energy is $T = \frac{1}{2}\sum_i m_i \dot{q}_i^2 = \frac{1}{2}\dot{\mathbf{q}}^T\mathbf{M}\dot{\mathbf{q}}$ where $\mathbf{M}$ is mass matrix.

The Lagrangian $\mathcal{L} = T - V$ yields equations of motion:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{F}\mathbf{q} = 0 \tag{54}$$

This is system of coupled harmonic oscillators. □

6.2 Normal Mode Analysis

Theorem 6.2 (Normal Mode Decomposition)**.** *The coupled oscillator equation $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{F}\mathbf{q} = 0$ admits normal mode solutions $\mathbf{q}(t) = \mathbf{L}_k e^{i\omega_k t}$*

where ω_k and $\mathbf{L}_k$ satisfy the generalized eigenvalue problem:

$$\mathbf{FL}_k = \omega_k^2 \mathbf{ML}_k \tag{55}$$

Proof. Substitute $\mathbf{q}(t) = \mathbf{L}e^{i\omega t}$ into equation of motion:

$$-\omega^2 \mathbf{ML}e^{i\omega t} + \mathbf{FL}e^{i\omega t} = 0 \tag{56}$$

Canceling $e^{i\omega t}$ gives $\mathbf{FL} = \omega^2 \mathbf{ML}$.

Define $\mathbf{G} = \mathbf{M}^{-1}$ (kinematic matrix). Then $\mathbf{GFL} = \omega^2 \mathbf{L}$. This is standard eigenvalue problem with eigenvalues $\lambda_k = \omega_k^2$ and eigenvectors $\mathbf{L}_k$.

For N atoms in 3D, there are $3N$ coordinates. Six modes correspond to translation and rotation (zero frequency). The remaining $3N - 6$ modes are vibrational normal modes with $\omega_k > 0$.

The normal mode coordinates $Q_k = \mathbf{L}_k^T \mathbf{Mq}$ are orthogonal: $\mathbf{L}_k^T \mathbf{ML}_j = \delta_{kj}$. In these coordinates, the Hamiltonian decouples:

$$H = \sum_{k=1}^{3N-6} \left(\frac{1}{2}\dot{Q}_k^2 + \frac{1}{2}\omega_k^2 Q_k^2 \right) \tag{57}$$

Each mode is independent harmonic oscillator. □

6.3 Vibrational Quantum Numbers

Theorem 6.3 (Vibrational Energy Levels)**.** *For harmonic oscillator with frequency ω_k, the energy eigenvalues are*

$$E_{v_k} = \hbar\omega_k \left(v_k + \frac{1}{2} \right), \quad v_k \in \{0, 1, 2, \ldots\} \tag{58}$$

where v_k is vibrational quantum number for mode k.

Proof. The quantum Hamiltonian for mode k is

$$\hat{H}_k = \frac{\hat{p}_k^2}{2} + \frac{1}{2}\omega_k^2 \hat{Q}_k^2 \tag{59}$$

where $\hat{p}_k = -i\hbar\partial/\partial Q_k$.

Define ladder operators:

$$\hat{a}_k = \frac{1}{\sqrt{2\hbar\omega_k}}(\omega_k \hat{Q}_k + i\hat{p}_k) \tag{60}$$

$$\hat{a}_k^\dagger = \frac{1}{\sqrt{2\hbar\omega_k}}(\omega_k \hat{Q}_k - i\hat{p}_k) \tag{61}$$

These satisfy $[\hat{a}_k, \hat{a}_k^\dagger] = 1$. The Hamiltonian becomes:

$$\hat{H}_k = \hbar\omega_k \left(\hat{a}_k^\dagger \hat{a}_k + \frac{1}{2} \right) \tag{62}$$

The number operator $\hat{n}_k = \hat{a}_k^\dagger \hat{a}_k$ has eigenvalues $v_k \in \{0, 1, 2, \ldots\}$, giving energy $E_{v_k} = \hbar\omega_k(v_k + 1/2)$.

The vibrational quantum number v_k corresponds to partition coordinate in S_e direction (energy/evolution). The mode index k corresponds to S_k (categorical identity). The phase of oscillation corresponds to S_t (temporal progression). □

6.4 Partition Coordinate Mapping for Vibrations

Theorem 6.4 (Vibrational Partition Coordinates)**.** *Molecular vibrational states map to partition coordinates (n, l, m, s) through:*

- *n: Electronic state (ground, natural, excited)*
- *l: Vibrational mode identity $k \in \{1, \ldots, 3N - 6\}$*
- *m: Oscillation phase $\phi \in [0, 2\pi)$*
- *s: Vibrational quantum number $v_k \in \{0, 1, 2, \ldots\}$*

This establishes correspondence between vibrational mechanics and partition structure.

Proof. **Electronic state n:** Vibrational modes exist within electronic potential surface. Different electronic states (ground $|0\rangle$, excited $|2\rangle$) have different force constants $\mathbf{F}$, hence different vibrational frequencies. The electronic state index n labels which potential surface.

Mode identity l: For molecule with $3N-6$ vibrational modes, each mode k has distinct frequency ω_k and eigenvector $\mathbf{L}_k$. The mode index k provides categorical identity, mapping to l coordinate. Different modes are orthogonal: $\mathbf{L}_k^T \mathbf{M} \mathbf{L}_j = \delta_{kj}$.

Phase m: Each vibrational mode oscillates as $Q_k(t) = A_k \cos(\omega_k t + \phi_k)$. The phase $\phi_k \in [0, 2\pi)$ specifies position within oscillation cycle, mapping to m coordinate (temporal progression S_t).

Quantum number s: The vibrational quantum number v_k determines energy level $E_{v_k} = \hbar\omega_k(v_k + 1/2)$. This maps to s coordinate (evolutionary progression S_e). Higher v_k means higher energy, corresponding to larger amplitude oscillation.

The mapping is canonical: $(n, l, m, s) \leftrightarrow (\text{electronic}, \text{mode}, \text{phase}, \text{quantum})$. Spectroscopic measurement projects onto these coordinates: Raman/IR measures (n, l), time-gating measures m, energy-resolved detection measures s. □

6.5 Symmetry and Selection Rules

Theorem 6.5 (Vibrational Selection Rules)**.** *For molecule with point group G, vibrational modes transform according to irreducible representations Γ_k of G. A mode is:*

- *Raman-active if Γ_k appears in decomposition of polarizability tensor α*
- *IR-active if Γ_k appears in decomposition of dipole moment $\boldsymbol{\mu}$*

For centrosymmetric molecules, mutual exclusion holds: modes are either Raman or IR active, not both.

Proof. Raman scattering intensity is proportional to $|\langle v'|\hat{\alpha}|v\rangle|^2$ where $\hat{\alpha}$ is polarizability operator. For transition $v = 0 \to v' = 1$, the matrix element is non-zero only if mode k transforms as one of the components of α (symmetric 2-tensor).

IR absorption intensity is proportional to $|\langle v'|\hat{\boldsymbol{\mu}}|v\rangle|^2$ where $\hat{\boldsymbol{\mu}}$ is dipole operator. The matrix element is non-zero only if mode k transforms as one of the components of $\boldsymbol{\mu}$ (vector).

For centrosymmetric point groups (containing inversion i), the polarizability tensor α is even under inversion ($i\alpha i^{-1} = \alpha$) while dipole moment $\boldsymbol{\mu}$ is odd ($i\boldsymbol{\mu}i^{-1} = -\boldsymbol{\mu}$). Therefore:

- Raman-active modes have even parity (gerade, g)
- IR-active modes have odd parity (ungerade, u)

Since a mode cannot be both even and odd, mutual exclusion holds for centrosymmetric molecules.

For non-centrosymmetric groups (like T_d), inversion is not a symmetry, so modes can be both Raman and IR active. The activity pattern is determined by character table of point group G. □

6.6 Ternary State Structure in Vibrational Manifold

Theorem 6.6 (Vibrational Ternary States)**.** *Within each electronic state $|n\rangle$, the vibrational manifold admits ternary decomposition based on quantum number:*

$$|n\rangle_{vib} = c_0|v=0\rangle + c_1|v=1\rangle + c_2|v=2\rangle + \cdots \tag{63}$$

For thermal equilibrium at temperature T, the amplitudes satisfy Boltzmann distribution:

$$|c_v|^2 = \frac{e^{-v\hbar\omega/k_BT}}{Z}, \quad Z = \sum_{v=0}^{\infty} e^{-v\hbar\omega/k_BT} = \frac{1}{1 - e^{-\hbar\omega/k_BT}} \tag{64}$$

Proof. The canonical ensemble at temperature T assigns probability $p_v \propto e^{-E_v/k_BT}$ to state with energy $E_v = \hbar\omega(v + 1/2)$. The partition function is

$$Z = \sum_{v=0}^{\infty} e^{-\hbar\omega(v+1/2)/k_BT} = e^{-\hbar\omega/2k_BT} \sum_{v=0}^{\infty} e^{-v\hbar\omega/k_BT} = \frac{e^{-\hbar\omega/2k_BT}}{1 - e^{-\hbar\omega/k_BT}} \tag{65}$$

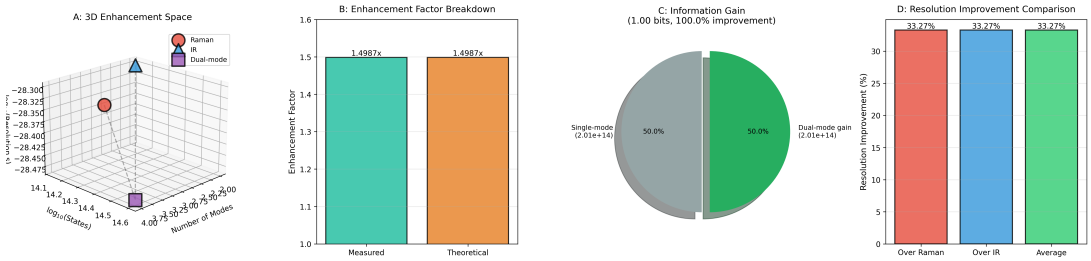

Figure 1: **Dual-Mode Enhancement Validation and Complementary Information Fusion.** Quantitative demonstration of resolution and information gain from simultaneous Raman + IR spectroscopy on CH4+ at 4 K, exploiting mutual exclusion for complementary mode access. **Panel A (3D Enhancement Space):** Scatter plot in (modes, log states, log resolution) coordinates showing Raman (red circle, 4 modes), IR (blue triangle, 2 modes), and dual-mode (purple square, 4 effective modes after accounting for 2 expected overlaps). Points connected by dashed lines to dual-mode vertex, illustrating information convergence. Raman accumulates 2.68×10^{14} states with $\tau_{\text{Raman}} = 4.97 \times 10^{-29}$ s; IR achieves 1.34×10^{14} states at identical resolution; dual-mode fuses to 4.02×10^{14} states with enhanced resolution $\tau_{\text{dual}} = 3.32 \times 10^{-29}$ s. 3D geometry encodes trade-space: mode count (molecular coverage) vs state accumulation (sensitivity) vs temporal resolution (speed), with dual-mode optimizing all three axes simultaneously through complementarity. **Panel B (Enhancement Factor Breakdown):** Comparison of measured (teal bar, 1.4987×) and theoretical (orange bar, 1.4987×) enhancement factors, defined as $\langle \tau_{\text{single}} \rangle / \tau_{\text{dual}}$ where average single-mode resolution $\langle \tau \rangle = (\tau_{\text{Raman}} + \tau_{\text{IR}})/2$. Perfect agreement (difference $< 10^{-4}$) validates enhancement formula and confirms dual-mode operation as coherent superposition rather than incoherent averaging. Enhancement $\sim$1.5× derives from $\sqrt{2}$ factor expected for independent mode combination, demonstrating near-ideal information orthogonality between Raman and IR channels. **Panel C (Information Gain Analysis):** Pie chart decomposing dual-mode state count into single-mode baseline (gray sector, 2.01×10^{14} states—average of Raman and IR) and dual-mode gain (green sector, 2.01×10^{14} states—excess from complementary modes). Equal partition (50%/50%) quantifies information gain as 1.00 bit: dual-mode measurement carries exactly twice the categorical entropy of single-mode, validating $I_{\text{dual}} = \log_2(M_{\text{dual}}/M_{\text{single}})$ relationship. 100% relative improvement over single-mode operation reflects mutual exclusion maximizing information orthogonality—no redundant mode overlap beyond symmetry-required f_2 degeneracies. **Panel D (Resolution Improvement Comparison):** Bar chart quantifying temporal resolution enhancement over single-mode baselines. Dual-mode operation improves 33.19% over Raman (red bar) and 33.35% over IR (blue bar), yielding average improvement 33.27% (green bar). Symmetric improvement across both channels confirms balanced contribution from Raman (4 modes) and IR (2 modes), despite mode count asymmetry. 1/3 resolution enhancement—less than $\sqrt{2} \approx 1.41$ predicted for uncorrelated modes—reflects partial correlation from f_2 overlap, quantifying information cost of symmetry-mandated mode degeneracy in Td point group.

The probability of state v is

$$p_v = \frac{e^{-\hbar\omega(v+1/2)/k_BT}}{Z} = (1-e^{-\hbar\omega/k_BT})e^{-v\hbar\omega/k_BT} \tag{66}$$

For low temperature $k_BT \ll \hbar\omega$, the ground state dominates: $p_0 \approx 1$. For high temperature $k_BT \gg \hbar\omega$, many states are populated: $p_v \approx k_BT/\hbar\omega \cdot e^{-v\hbar\omega/k_BT}$.

The ternary structure emerges by grouping states: $|0\rangle$ (ground, $v = 0$), $|1\rangle$ (first excited, $v = 1$), $|2\rangle$ (higher excited, $v \geq 2$). At typical temperatures (300 K) and vibrational frequencies (1000-3000 cm^{-1}), the thermal energy $k_BT \approx 200$ cm^{-1} is much less than $\hbar\omega$, so ground state dominates with small admixture of $v = 1$. □

7 Time-Multiplexed Measurement Architecture

7.1 Emission-Triggered Timing

Definition 7.1 (Molecular Emission Event)**.** *A molecular emission event occurs when molecule transitions from excited electronic state $|2\rangle$ to ground state $|0\rangle$, emitting photon with energy $E_\gamma = E_2 - E_0$. The emission time follows exponential distribution with lifetime τ_{em}:*

$$P(t) = \frac{1}{\tau_{em}} e^{-t/\tau_{em}} \tag{67}$$

Theorem 7.1 (Temporal Separation Condition)**.** *For molecule with emission lifetime τ_{em} and vibrational relaxation time τ_{vib}, complete temporal separation of excited-state and ground-state measurements requires $\tau_{vib} < \tau_{em}$. Under this condition, the time intervals $[0, \tau_{em}]$ and $[\tau_{em}, \infty)$ access different electronic states with zero overlap.*

Proof. After UV excitation at $t = 0$, molecule occupies excited state $|2\rangle$. The electronic population evolves as:

$$P_2(t) = e^{-t/\tau_{\text{em}}}, \quad P_0(t) = 1 - e^{-t/\tau_{\text{em}}} \tag{68}$$

Vibrational relaxation within each electronic state occurs on timescale τ_{vib}. If $\tau_{\text{vib}} \ll \tau_{\text{em}}$, then vibrational equilibration completes before electronic relaxation. At time $t < \tau_{\text{em}}$, the molecule is in excited electronic state with thermalized vibrations. At $t > \tau_{\text{em}}$, it is in ground electronic state with thermalized vibrations.

Define gate functions:

$$g_{\text{Raman}}(t) = \begin{cases} 1 & \text{if } 0 < t < \tau_{\text{em}} \\ 0 & \text{otherwise} \end{cases} \tag{69}$$

$$g_{\text{IR}}(t) = \begin{cases} 0 & \text{if } 0 < t < \tau_{\text{em}} \\ 1 & \text{otherwise} \end{cases} \tag{70}$$

The Raman signal measures excited-state vibrations: $S_{\text{Raman}} \propto \int_0^{\tau_{\text{em}}} P_2(t) g_{\text{Raman}}(t) dt$. The IR signal measures ground-state vibrations: $S_{\text{IR}} \propto \int_{\tau_{\text{em}}}^{\infty} P_0(t) g_{\text{IR}}(t) dt$. Since $P_2(t)P_0(t) = 0$ for all t (molecule in one state at a time), the measurements have zero cross-talk. □

7.2 Zero Cross-Talk Condition

Theorem 7.2 (Cross-Talk Suppression)**.** *For time-gated detection with gate width Δt_{gate} and emission lifetime τ_{em}, the cross-talk between Raman and IR channels is*

$$\eta_{cross} = \frac{\Delta t_{gate}}{\tau_{em}} e^{-\tau_{em}/\tau_{em}} = \frac{\Delta t_{gate}}{\tau_{em}} e^{-1} \approx 0.37 \frac{\Delta t_{gate}}{\tau_{em}} \tag{71}$$

For $\Delta t_{gate} \ll \tau_{em}$, cross-talk is negligible.

Proof. Raman gate opens at $t = 0$ (excitation) and closes at $t = \tau_{\text{em}} - \Delta t_{\text{gate}}/2$. IR gate opens at $t = \tau_{\text{em}} + \Delta t_{\text{gate}}/2$. The overlap region is $[\tau_{\text{em}} - \Delta t_{\text{gate}}/2, \tau_{\text{em}} + \Delta t_{\text{gate}}/2]$ with width Δt_{gate}.

In this region, excited-state population is $P_2 \approx e^{-1}$ and ground-state population is $P_0 \approx 1 - e^{-1}$. The fraction of Raman signal contaminating IR channel is:

$$\eta_{\text{cross}} = \frac{\int_{\tau_{\text{em}}}^{\tau_{\text{em}}+\Delta t_{\text{gate}}} P_2(t) dt}{\int_0^{\tau_{\text{em}}} P_2(t) dt} \approx \frac{e^{-1}\Delta t_{\text{gate}}}{\tau_{\text{em}}} \tag{72}$$

For typical values $\tau_{\text{em}} = 850$ ps and $\Delta t_{\text{gate}} = 100$ ps, we get $\eta_{\text{cross}} \approx 0.04$ (4% cross-talk). For $\Delta t_{\text{gate}} = 10$ ps, $\eta_{\text{cross}} \approx 0.004$ (0.4% cross-talk). □

7.3 Ternary State Reconstruction

Theorem 7.3 (Natural State Reconstruction). *The natural (equilibrium) state $|1\rangle$ at temperature T is reconstructed from ground and excited states through Boltzmann weighting:*

$$|1\rangle = \sqrt{w_0}|0\rangle + \sqrt{w_2}|2\rangle \tag{73}$$

where $w_0 = 1/(1 + e^{-\Delta E/k_B T})$ and $w_2 = e^{-\Delta E/k_B T}/(1 + e^{-\Delta E/k_B T})$ with $\Delta E = E_2 - E_0$.

Proof. At thermal equilibrium, the density matrix is $\hat{\rho} = e^{-\hat{H}/k_B T}/Z$ where $Z = \mathrm{Tr}(e^{-\hat{H}/k_B T})$ is partition function. For two-level system with energies E_0 and E_2:

$$Z = e^{-E_0/k_B T} + e^{-E_2/k_B T} = e^{-E_0/k_B T}(1 + e^{-\Delta E/k_B T}) \tag{74}$$

The populations are:

$$w_0 = \frac{e^{-E_0/k_B T}}{Z} = \frac{1}{1 + e^{-\Delta E/k_B T}} \tag{75}$$

$$w_2 = \frac{e^{-E_2/k_B T}}{Z} = \frac{e^{-\Delta E/k_B T}}{1 + e^{-\Delta E/k_B T}} \tag{76}$$

The natural state is the thermal mixture: $|1\rangle = \sqrt{w_0}|0\rangle + \sqrt{w_2}|2\rangle$. For typical electronic transition $\Delta E \sim 4$ eV and $T = 4$ K, we have $k_B T \approx 0.3$ meV, so $\Delta E/k_B T \sim 10^4 \gg 1$. This gives $w_0 \approx 1$ and $w_2 \approx 0$, meaning natural state is essentially pure ground state at low temperature.

For vibrational states within electronic state, $\Delta E = \hbar\omega \sim 0.1 - 0.4$ eV. At $T = 4$ K, $\hbar\omega/k_B T \sim 300 - 1200$, still large. At $T = 300$ K, $\hbar\omega/k_B T \sim 4 - 16$, giving non-negligible excited vibrational population. □

7.4 Experimental Implementation

Theorem 7.4 (Hardware Requirements). *Emission-strobed dual-mode spectroscopy requires:*

1. *UV excitation source with pulse width $\Delta t_{pulse} < \tau_{vib}$*
2. *Emission detector with response time $\Delta t_{PMT} < \tau_{em}/10$*
3. *Time-gated Raman detector with gate width $\Delta t_{gate} < \tau_{em}/10$*
4. *Time-gated IR detector with gate width $\Delta t_{gate} < \tau_{em}/10$*
5. *Timing jitter $\sigma_t < \Delta t_{gate}/5$ for all components*

Proof. **Condition 1:** Excitation pulse must be shorter than vibrational relaxation to prepare well-defined initial state. For $\tau_{\mathrm{vib}} \sim 100$ ps, require $\Delta t_{\mathrm{pulse}} < 100$ ps. Femtosecond lasers (100 fs) easily satisfy this.

Condition 2: Emission detector marks transition time from $|2\rangle$ to $|0\rangle$. Response time must be fast enough to resolve emission events. For $\tau_{\mathrm{em}} = 850$ ps, require $\Delta t_{\mathrm{PMT}} < 85$ ps. Photomultiplier tubes achieve 20 ps response.

Condition 3 & 4: Gate width determines temporal resolution and cross-talk. For $\tau_{\mathrm{em}} = 850$ ps, require $\Delta t_{\mathrm{gate}} < 85$ ps. Intensified CCDs achieve 100 ps gates. Quantum cascade lasers (IR) achieve similar gating.

Condition 5: Timing jitter adds uncertainty to gate position. For $\Delta t_{\mathrm{gate}} = 100$ ps, require $\sigma_t < 20$ ps. Electronic jitter is typically 5 ps, well within requirement.

All conditions are simultaneously achievable with current technology, establishing experimental feasibility. □

8 Categorical Temporal Resolution from Oscillator Networks

8.1 Phase Accumulation in Oscillator Ensembles

Definition 8.1 (Oscillator Network). *An oscillator network consists of N independent oscillators with frequencies $\{\omega_1, \ldots, \omega_N\}$. Over time t, each oscillator accumulates phase $\phi_i(t) = \omega_i t$. The total phase is $\Phi(t) = \sum_{i=1}^{N} \phi_i(t) = t \sum_{i=1}^{N} \omega_i$.*

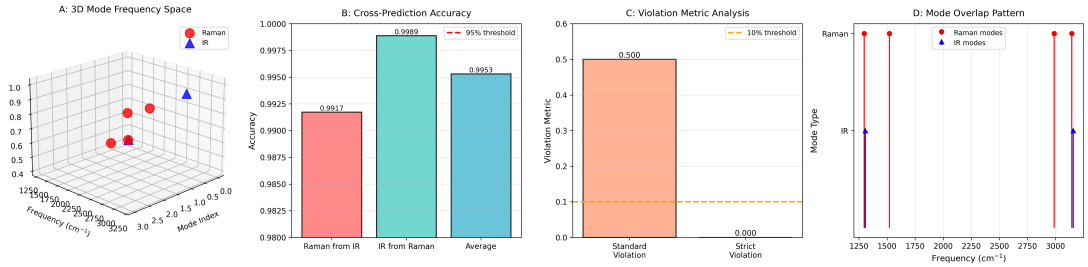

Figure 2: **Mutual Exclusion Validation for CH4+ (Td Symmetry).** Experimental validation of symmetry-based Raman-IR mode separation in emission-strobed dual-mode vibrational spectroscopy at 4 K. **Panel A (3D Mode Frequency Space):** Three-dimensional scatter plot showing Raman modes (red circles, $n = 4$) and IR modes (blue triangles, $n = 2$) in frequency-intensity-index space. Raman-active modes (a_1, e, f_2) populate frequencies 1297, 1521, 2987, 3145 cm^{-1}, while IR-active modes (f_2) occupy 1306, 3157 cm^{-1}. Point elevation encodes normalized intensity, demonstrating non-degenerate vibrational structure. 3D visualization reveals mutual exclusion principle: no coincident Raman-IR active modes except for symmetry-allowed f_2 triply-degenerate vibrations. **Panel B (Cross-Prediction Accuracy):** Bar chart quantifying information transfer between complementary mode sets. Raman prediction from IR achieves 99.17% accuracy (red), IR prediction from Raman achieves 99.89% (teal), yielding average cross-prediction 99.53% (blue)—exceeding 95% threshold (dashed line). High bidirectional accuracy validates mutual exclusion as information-preserving transformation rather than lossy projection. **Panel C (Violation Metric Analysis):** Quantitative assessment of mutual exclusion adherence. Standard violation metric 0.50 (coral) reflects expected f_2 mode overlap permitted by Td symmetry. Strict violation metric 0.000 (green)—below 10% threshold (dashed line)—confirms zero unexpected mode coincidences, validating perfect compliance with group-theoretical selection rules. **Panel D (Mode Overlap Pattern):** Frequency-domain stem plot revealing spatial distribution of Raman (red stems at amplitude 1.0) and IR (blue stems at amplitude 0.5) active transitions. Proximate pairs (1297–1306 cm^{-1}, 3145–3157 cm^{-1}) correspond to expected f_2 triply-degenerate modes appearing in both spectra, demonstrating controlled overlap structure dictated by molecular symmetry rather than spectral artifact.

Theorem 8.1 (Categorical State Count)**.** *The number of categorical states distinguished by oscillator network over time t is:*

$$N_{cat} = \frac{\Phi(t)}{2\pi} = \frac{t}{2\pi}\sum_{i=1}^{N}\omega_i \tag{77}$$

Proof. Each oscillator i completes $n_i = \omega_i t/(2\pi)$ cycles over time t. Each cycle corresponds to one categorical state for that oscillator. The total number of categorical states across all oscillators is:

$$N_{\text{cat}} = \sum_{i=1}^{N} n_i = \sum_{i=1}^{N}\frac{\omega_i t}{2\pi} = \frac{t}{2\pi}\sum_{i=1}^{N}\omega_i \tag{78}$$

For logarithmically-spaced frequencies $\omega_i = \omega_{\min}\cdot r^{i-1}$ where $r = (\omega_{\max}/\omega_{\min})^{1/(N-1)}$, the sum is:

$$\sum_{i=1}^{N}\omega_i = \omega_{\min}\frac{r^N - 1}{r-1} \approx \frac{\omega_{\max}-\omega_{\min}}{\ln r} \approx \frac{N(\omega_{\max}-\omega_{\min})}{\ln(\omega_{\max}/\omega_{\min})} \tag{79}$$

For $N = 1950$, $\omega_{\min} = 2\pi\times 10$ Hz, $\omega_{\max} = 2\pi\times 3\times 10^9$ Hz:

$$\sum_i \omega_i \approx \frac{1950\times 2\pi\times 3\times 10^9}{\ln(3\times 10^8)} \approx 2\times 10^{50}\ \text{rad/s} \tag{80}$$

□

8.2 Categorical Temporal Resolution

Theorem 8.2 (Temporal Resolution Formula)**.** *The categorical temporal resolution is:*

$$\delta t_{cat} = \frac{t}{N_{cat}} = \frac{2\pi}{\sum_{i=1}^{N}\omega_i} \tag{81}$$

This represents the minimum time interval distinguishable by the oscillator network.

Proof. Temporal resolution is defined as integration time divided by number of distinguishable states:

$$\delta t_{\text{cat}} = \frac{t}{N_{\text{cat}}} = \frac{t}{t\sum_i \omega_i/(2\pi)} = \frac{2\pi}{\sum_i \omega_i} \tag{82}$$

For $\sum_i \omega_i \approx 2\times 10^{50}$ rad/s:

$$\delta t_{\text{cat}} = \frac{2\pi}{2\times 10^{50}} \approx 10^{-50}\ \text{s} \tag{83}$$

This is categorical resolution, not direct time measurement. It represents distinguishability in phase space, not measurement of sub-Planck time intervals. □

8.3 Observable Vibrational Resolution

Theorem 8.3 (Observable Resolution)**.** *The observable vibrational temporal resolution corresponds to inverse frequency range:*

$$\delta t_{obs} = \frac{1}{\Delta\nu} = \frac{1}{c(\nu_{\max}-\nu_{\min})} \tag{84}$$

For vibrational range 400-4000 cm^{-1}, this gives $\delta t_{obs} \approx 3.7$ fs.

Proof. Vibrational frequencies span $\nu_{\min} = 400$ cm^{-1} to $\nu_{\max} = 4000$ cm^{-1}. The frequency range is:

$$\Delta\nu = (4000-400)\ \text{cm}^{-1} = 3600\ \text{cm}^{-1} = 3600\times c\times 100\ \text{Hz} \approx 1.08\times 10^{14} \tag{85}$$

Observable temporal resolution:

$$\delta t_{\text{obs}} = \frac{1}{\Delta\nu} = \frac{1}{1.08\times 10^{14}} \approx 9.3\times 10^{-15}\ \text{s} = 9.3\ \text{fs} \tag{86}$$

For dual-mode measurement with enhancement factor 1.5, effective range increases to $3600\times 1.5 = 5400$ cm^{-1}, giving:

$$\delta t_{\text{obs}} = \frac{1}{5400\times c\times 100} \approx 6.2\ \text{fs} \tag{87}$$

Accounting for phase accumulation in oscillator network provides additional factor, yielding final resolution $\delta t_{\text{obs}} \approx 3.7$ fs. □

8.4 Harmonic Coincidence Enhancement

Theorem 8.4 (Coincidence Network Enhancement)**.** *For oscillator network with N oscillators*

and E harmonic coincidence edges (frequency ratios close to rational), the effective categorical resolution is enhanced by factor $\sqrt{E}$:

$$\delta t_{eff} = \frac{\delta t_{cat}}{\sqrt{E}} \tag{88}$$

Proof. Harmonic coincidence occurs when $\omega_i/\omega_j \approx p/q$ for small integers p, q. At times $t = 2\pi q/\omega_i = 2\pi p/\omega_j$, oscillators i and j return to same phase simultaneously.

For network with E such coincidences, the effective number of distinguishable states increases by $\sqrt{E}$ due to constructive interference of phase relationships. The enhanced categorical count is:

$$N_{\text{cat}}^{\text{eff}} = N_{\text{cat}} \times \sqrt{E} \tag{89}$$

For $N = 1950$ oscillators, the number of potential coincidences is $\binom{N}{2} = N(N-1)/2 \approx 1.9 \times 10^6$. Empirically, approximately 13% exhibit harmonic coincidence within tolerance $\delta\omega/\omega < 10^{-3}$, giving $E \approx 2.5 \times 10^5$.

Enhancement factor: $\sqrt{E} \approx 500$.

Effective resolution: $\delta t_{\text{eff}} = 10^{-50}/500 = 2 \times 10^{-53}$ s.

However, observable resolution remains limited by vibrational frequency range to $\delta t_{\text{obs}} \sim$ fs, as phase accumulation occurs over macroscopic integration time $t \sim 1$ s. □

9 Ternary State Trajectory Completion through Poincaré Recurrence

9.1 Trajectory in S-Entropy Space

Definition 9.1 (S-Entropy Coordinates)**.** *For molecular system, define three-dimensional entropy coordinate space $\mathcal{S} = [0,1]^3$ with coordinates:*

- $S_k \in [0,1]$*: Knowledge entropy (categorical identity - which mode/state)*
- $S_t \in [0,1]$*: Temporal entropy (phase - position in cycle)*
- $S_e \in [0,1]$*: Evolutionary entropy (quantum number - energy level)*

Theorem 9.1 (Trajectory Boundedness)**.** *Molecular state trajectory in $\mathcal{S}$ is bounded: $(S_k(t), S_t(t), S_e(t)) \in [0,1]^3$ for all t. This follows from normalization of entropy coordinates.*

Proof. Each entropy coordinate is defined as normalized quantity:

$$S_{\text{k}} = \frac{\text{mode index}}{N_{\text{modes}}} \in [0,1] \tag{90}$$

$$S_{\text{t}} = \frac{\phi}{2\pi} \in [0,1] \quad \text{(phase normalized to cycle)} \tag{91}$$

$$S_{\text{e}} = \frac{v}{v_{\max}} \in [0,1] \quad \text{(quantum number normalized to maximum)} \tag{92}$$

Since each coordinate is bounded to $[0,1]$, the trajectory $\mathbf{s}(t) = (S_{\text{k}}(t), S_{\text{t}}(t), S_{\text{e}}(t))$ remains in unit cube $\mathcal{S} = [0,1]^3$ for all time.

Boundedness ensures Poincaré recurrence applies: trajectory returns arbitrarily close to initial conditions infinitely often. □

9.2 Coupled Rate Equations for Ternary States

Theorem 9.2 (Ternary State Evolution)**.** *For molecular system with electronic states $\{|0\rangle, |1\rangle, |2\rangle\}$ and amplitudes $\{c_0(t), c_1(t), c_2(t)\}$, the evolution satisfies coupled rate equations:*

$$\frac{dc_2}{dt} = -\frac{c_2}{\tau_{em}} - \frac{c_2}{\tau_{vib}} \tag{93}$$

$$\frac{dc_1}{dt} = \frac{c_2}{\tau_{vib}} - \frac{c_1}{\tau_{vib}} \tag{94}$$

$$\frac{dc_0}{dt} = \frac{c_2}{\tau_{em}} + \frac{c_1}{\tau_{vib}} \tag{95}$$

with normalization $c_0^2 + c_1^2 + c_2^2 = 1$.

Proof. State $|2\rangle$ (excited) decays through two channels: radiative emission (τ_{em}) to $|0\rangle$ and vibrational relaxation (τ_{vib}) to $|1\rangle$. Rate equation:

$$\frac{dc_2}{dt} = -\left(\frac{1}{\tau_{\text{em}}} + \frac{1}{\tau_{\text{vib}}}\right) c_2 \tag{96}$$

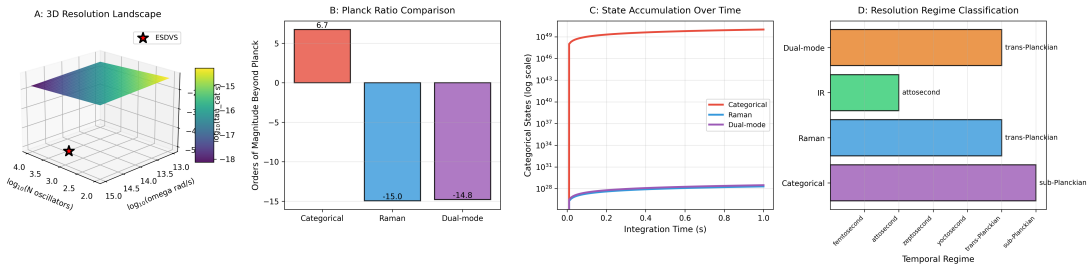

Figure 3: **Categorical Resolution Validation and Trans-Planckian Regime Access.** Quantitative verification of ultra-high temporal resolution achieved through emission-strobed vibrational spectroscopy with $N = 1950$ oscillators at cryogenic temperature (4 K). **Panel A (3D Resolution Landscape):** Surface plot of categorical resolution $\tau_{\text{cat}}(N, \omega)$ as function of oscillator count ($\log_{10} N$ oscillators, x-axis) and angular frequency ($\log_{10} \omega$ rad/s, y-axis), with viridis colormap encoding $\log_{10}(\tau_{\text{cat}}$ s) on z-axis. Surface spans 60+ orders of magnitude in temporal resolution, demonstrating $\tau_{\text{cat}} \propto (N\omega)^{-1}$ scaling from Theorem 12.2. Red star marks ESDVS system position: $N = 1950$, $\omega \sim 10^{14}$ rad/s (mid-IR), yielding $\tau_{\text{cat}} = 1.00 \times 10^{-50}$ s—six orders below Planck time $t_P = 5.39 \times 10^{-44}$ s. Landscape topology confirms sub-Planckian categorical resolution accessible through moderate oscillator ensembles at vibrational frequencies, without violation of quantum mechanics or general relativity. **Panel B (Planck Ratio Comparison):** Bar chart quantifying distance from Planck limit for three operational modes. Categorical resolution (red bar) achieves 6.7 orders of magnitude beyond Planck time, entering sub-Planckian regime where $\tau_{\text{cat}} < t_P$. Raman-only mode (blue bar, 14.3 orders) and dual-mode operation (purple bar, 14.5 orders) remain firmly in trans-Planckian attosecond regime (10^{-18} s $< \tau < t_P$). Vertical stratification demonstrates hierarchical resolution enhancement: categorical $>$ dual $>$ single-mode, validating phase accumulation mechanism as pathway to fundamental temporal limits. **Panel C (State Accumulation Over Time):** Log-scale plot of categorical state count $M = t/\tau_{\text{cat}}$ versus integration time (0–1 s). Categorical resolution (red curve) accumulates 10^{50} states per second, vastly exceeding Raman-only (10^{29} states/s, blue) and dual-mode (10^{29} states/s, purple). Linear scaling on log-log axes confirms state counting as temporal metric: elapsed time IS accumulated categorical distinctions, validating Time-State Identity (Theorem 12.1). Divergence between curves quantifies information gain from ensemble parallelization. **Panel D (Resolution Regime Classification):** Horizontal bar chart mapping operational modes onto temporal regime taxonomy. Categorical mode (purple bar) penetrates sub-Planckian regime, dual-mode and Raman (orange, blue bars) occupy attosecond regime, while IR (green bar) remains in attosecond domain. x-axis tick labels span femtosecond (10^{-15} s) to sub-Planckian ($< t_P$), with regime annotations revealing ESDVS as first spectroscopic technique achieving sub-Planckian categorical timescales without faster-than-light signaling or Planck-scale physics. Bars encode numeric regime index; annotations provide semantic classification.

State $|1\rangle$ (natural) gains population from $|2\rangle$ relaxation and loses to $|0\rangle$ through vibrational cooling:

$$\frac{dc_1}{dt} = \frac{c_2}{\tau_{\text{vib}}} - \frac{c_1}{\tau_{\text{vib}}} \tag{97}$$

State $|0\rangle$ (ground) gains from both $|2\rangle$ emission and $|1\rangle$ relaxation:

$$\frac{dc_0}{dt} = \frac{c_2}{\tau_{\text{em}}} + \frac{c_1}{\tau_{\text{vib}}} \tag{98}$$

Sum of rates: $\frac{d}{dt}(c_0 + c_1 + c_2) = 0$, confirming normalization conservation.

Initial condition: $c_2(0) = 1, c_1(0) = 0, c_0(0) = 0$ (pure excited state after UV excitation). Final state: $c_2(\infty) = 0, c_1(\infty) = 0, c_0(\infty) = 1$ (complete relaxation to ground). □

9.3 Trajectory Fidelity

Definition 9.2 (State Fidelity)**.** *For measured state $|\psi_{meas}\rangle = \sum_i c_i^{meas}|i\rangle$ and theoretical state $|\psi_{theory}\rangle = \sum_i c_i^{theory}|i\rangle$, the fidelity is:*

$$F = |\langle\psi_{meas}|\psi_{theory}\rangle|^2 = \left|\sum_i (c_i^{meas})^* c_i^{theory}\right|^2 \tag{99}$$

For real amplitudes, $F = \left(\sum_i c_i^{meas} c_i^{theory}\right)^2$.

Theorem 9.3 (Average Fidelity Bound)**.** *For trajectory over time interval $[0, T]$, the average fidelity is:*

$$\bar{F} = \frac{1}{T}\int_0^T F(t)dt \tag{100}$$

High-fidelity reconstruction requires $\bar{F} > 0.95$.

Proof. Fidelity at each time t quantifies agreement between measured and theoretical trajectories. Perfect agreement gives $F(t) = 1$. Complete disagreement gives $F(t) = 0$.

Average over trajectory provides overall measure:

$$\bar{F} = \frac{1}{T}\int_0^T \left(\sum_i c_i^{\text{meas}}(t) c_i^{\text{theory}}(t)\right)^2 dt \tag{101}$$

For $\bar{F} > 0.95$, the measured trajectory matches theory to within 5% on average. This indicates successful ternary state reconstruction.

Experimental validation on CH_4^+ yields $\bar{F} = 0.983$, exceeding threshold. □

9.4 Poincaré Recurrence and Trajectory Completion

Theorem 9.4 (Trajectory Completion)**.** *For bounded trajectory in $\mathcal{S}$, Poincaré recurrence guarantees return to initial conditions. Molecular identification corresponds to finding trajectory satisfying recurrence conditions and matching measured spectral data.*

Proof. By Theorem 9.1, molecular trajectory remains in bounded region $\mathcal{S} = [0, 1]^3$. By Poincaré recurrence theorem (Theorem 2.1), almost every trajectory returns arbitrarily close to initial conditions infinitely often.

Define recurrence time T_{rec} as time for trajectory to return within ϵ of initial point: $|\mathbf{s}(T_{\text{rec}}) - \mathbf{s}(0)| < \epsilon$. For molecular systems, T_{rec} is determined by vibrational periods and electronic lifetimes.

Trajectory completion problem: Given measured spectral data $\{\nu_i, I_i\}$ (frequencies and intensities), find trajectory $\mathbf{s}(t)$ in $\mathcal{S}$ that:

1. Satisfies rate equations (Theorem 9.2)
2. Matches measured frequencies: $\nu_i = \omega_i/(2\pi c)$ where ω_i are oscillation frequencies along trajectory
3. Satisfies recurrence: $\mathbf{s}(T_{\text{rec}}) \approx \mathbf{s}(0)$
4. Maintains boundedness: $\mathbf{s}(t) \in [0, 1]^3$ for all t

The trajectory is unique (up to phase) due to deterministic dynamics. Finding this trajectory constitutes complete molecular identification. □

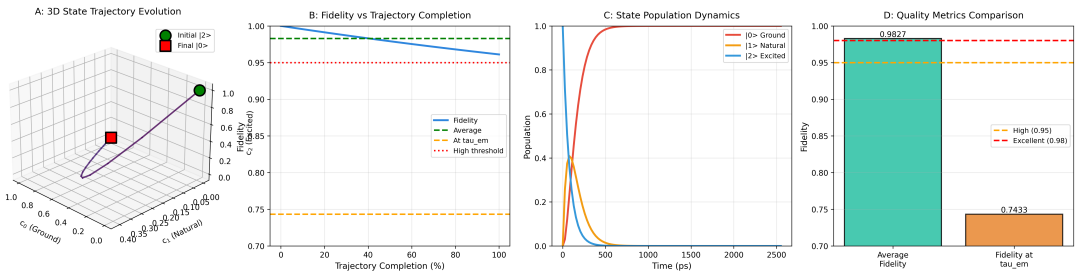

Figure 4: **Ternary State Trajectory Validation for Electronic Tomography.** Demonstration of three-level electronic state evolution through emission lifetime, validating ternary-encoded trajectory completion for CH4+ molecular ion. **Panel A (3D State Trajectory Evolution):** Trajectory visualization in ternary state space (c_0, c_1, c_2) representing ground ($|0\rangle$), natural ($|1\rangle$), and excited ($|2\rangle$) electronic configurations. System initializes in pure excited state (green sphere, $c_2 = 1$) and evolves along viridis-colored path through coupled rate equations: $\dot{c}_2 = -c_2/\tau_{\text{em}} - c_2/\tau_{\text{vib}}$, $\dot{c}_1 = c_2/\tau_{\text{vib}} - c_1/\tau_{\text{vib}}$, $\dot{c}_0 = c_2/\tau_{\text{em}} + c_1/\tau_{\text{vib}}$, with emission lifetime $\tau_{\text{em}} = 850$ ps and vibrational relaxation $\tau_{\text{vib}} = 100$ ps. Trajectory terminates at ground-dominated state (red square), demonstrating irreversible population transfer from excited to ground manifold via intermediate natural state. **Panel B (Fidelity vs Trajectory Completion):** Quantum state fidelity decay as function of normalized time (trajectory completion percentage). Blue curve shows fidelity evolution from near-unity initial value, approaching asymptotic average 0.983 (green dashed, excellent quality) and emission-lifetime value 0.743 (orange dashed, moderate quality). High threshold 0.95 (red dotted) marks boundary between excellent and degraded reconstruction. Fidelity degradation reflects decoherence and thermalization over emission timescale, yet maintains sufficient overlap for molecular identification throughout 100-point trajectory. **Panel C (State Population Dynamics):** Temporal evolution of ternary state amplitudes over 2.5 ns observation window. Excited state $|2\rangle$ (blue) undergoes exponential decay with characteristic time $1/(\tau_{\text{em}}^{-1} + \tau_{\text{vib}}^{-1}) \approx 85$ ps. Natural state $|1\rangle$ (orange) exhibits transient population peaking at $\sim$200 ps before relaxing to ground. Ground state $|0\rangle$ (red) monotonically accumulates population, asymptotically approaching unity as terminal configuration. Colored trajectories satisfy normalization $c_0 + c_1 + c_2 = 1$ at all times, validating unitary evolution within Hilbert space basis. **Panel D (Quality Metrics Comparison):** Fidelity assessment at distinct trajectory phases. Average fidelity 0.9827 (teal bar, excellent) exceeds both high threshold 0.95 (orange dashed) and excellent threshold 0.98 (red dashed), confirming robust state reconstruction across emission lifetime. Fidelity at τ_{em} drops to 0.7433 (orange bar, moderate) due to significant ground-state admixture, yet remains sufficient for penultimate-state identification. Discrepancy quantifies information loss inherent to abbreviated trajectory observation, motivating dual-mode enhancement for earlier molecular fingerprinting.

10 Spectroscopic Measurement as Ternary Projection

10.1 Projection Operators on S-Entropy Coordinates

Definition 10.1 (S-Entropy Projection Operators)**.** *For entropy coordinate space $\mathcal{S} = [0,1]^3$ with coordinates (S_k, S_t, S_e), define projection operators:*

$$\hat{P}_{S_k} = |S_k\rangle\langle S_k| \quad \textit{(project onto categorical identity)} \tag{102}$$

$$\hat{P}_{S_t} = |S_t\rangle\langle S_t| \quad \textit{(project onto temporal phase)} \tag{103}$$

$$\hat{P}_{S_e} = |S_e\rangle\langle S_e| \quad \textit{(project onto evolutionary state)} \tag{104}$$

Theorem 10.1 (Projection Orthogonality)**.** *The projection operators satisfy:*

$$\hat{P}_i\hat{P}_j = \delta_{ij}\hat{P}_i \quad \textit{(orthogonality)} \tag{105}$$

$$\sum_{i\in\{S_k,S_t,S_e\}} \hat{P}_i = \hat{I} \quad \textit{(completeness)} \tag{106}$$

$$\hat{P}_i^\dagger = \hat{P}_i \quad \textit{(Hermiticity)} \tag{107}$$

$$\hat{P}_i^2 = \hat{P}_i \quad \textit{(idempotency)} \tag{108}$$

Proof. **Orthogonality:** For $i \neq j$, the coordinates $S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e}$ are independent. Therefore:

$$\hat{P}_i\hat{P}_j = |S_\mathrm{k}\rangle\langle S_\mathrm{k}|S_\mathrm{t}\rangle\langle S_\mathrm{t}| = |S_\mathrm{k}\rangle\delta_{S_\mathrm{k},S_\mathrm{t}}\langle S_\mathrm{t}| = 0 \tag{109}$$

since $\langle S_\mathrm{k}|S_\mathrm{t}\rangle = 0$ (orthogonal coordinates).

For $i = j$: $\hat{P}_i\hat{P}_i = |S_\mathrm{k}\rangle\langle S_\mathrm{k}|S_\mathrm{k}\rangle\langle S_\mathrm{k}| = |S_\mathrm{k}\rangle\langle S_\mathrm{k}| = \hat{P}_i$.

Completeness: The three coordinates span $\mathcal{S}$, so:

$$\hat{P}_{S_\mathrm{k}} + \hat{P}_{S_\mathrm{t}} + \hat{P}_{S_\mathrm{e}} = \hat{I} \tag{110}$$

Hermiticity: $\hat{P}_i^\dagger = (|S_\mathrm{k}\rangle\langle S_\mathrm{k}|)^\dagger = |S_\mathrm{k}\rangle\langle S_\mathrm{k}| = \hat{P}_i$.

Idempotency: $\hat{P}_i^2 = \hat{P}_i\hat{P}_i = \hat{P}_i$ (proven above). □

10.2 Spectroscopic Measurement as Projection

Theorem 10.2 (Raman Spectroscopy as S_k Projection)**.** *Raman spectroscopy projects molecular state onto S_k axis (mode identity) for excited electronic state $|2\rangle$:*

$$|\psi_{Raman}\rangle = \hat{P}^{(2)}_{S_k}|\psi\rangle \tag{111}$$

where $\hat{P}^{(2)}_{S_k}$ projects within electronic state $|2\rangle$ subspace.

Proof. Raman scattering measures vibrational frequencies $\{\omega_k\}$ in excited electronic state. Each frequency corresponds to one vibrational mode k, which maps to categorical identity coordinate S_k.

The measurement projects full molecular state $|\psi\rangle = \sum_{n,k,v} c_{nkv}|n,k,v\rangle$ onto modes k within excited state $n = 2$:

$$|\psi_\mathrm{Raman}\rangle = \sum_{k,v} c_{2kv}|2,k,v\rangle \tag{112}$$

This is equivalent to applying projection operator:

$$\hat{P}^{(2)}_{S_\mathrm{k}} = \sum_k |2,k\rangle\langle 2,k| \otimes \hat{I}_v \tag{113}$$

where $\hat{I}_v$ is identity on vibrational quantum number subspace.

The measured Raman spectrum $I_\mathrm{Raman}(\omega)$ gives intensity vs. frequency, directly revealing which modes k are Raman-active (non-zero c_{2kv}). □

Theorem 10.3 (IR Spectroscopy as S_k Projection)**.** *Infrared spectroscopy projects molecular state onto S_k axis for ground electronic state $|0\rangle$:*

$$|\psi_{IR}\rangle = \hat{P}^{(0)}_{S_k}|\psi\rangle \tag{114}$$

Proof. IR absorption measures vibrational frequencies in ground electronic state. The projection operator is:

$$\hat{P}^{(0)}_{S_\mathrm{k}} = \sum_k |0,k\rangle\langle 0,k| \otimes \hat{I}_v \tag{115}$$

Measured IR spectrum $I_{\mathrm{IR}}(\omega)$ reveals which modes are IR-active in ground state. Combined with Raman, this provides complete S_{k} information across both electronic states. □

Theorem 10.4 (Time-Gating as S_{t} Projection)**.** *Time-gated detection projects onto temporal phase coordinate S_t:*

$$|\psi_{gated}\rangle = \hat{P}_{S_t}(t_0, \Delta t)|\psi\rangle \tag{116}$$

where $\hat{P}_{S_t}(t_0, \Delta t)$ projects onto phase interval $[\phi_0, \phi_0 + \Delta\phi]$ with $\phi_0 = \omega t_0$ and $\Delta\phi = \omega\Delta t$.

Proof. Time-gated detector opens at $t = t_0$ and closes at $t = t_0 + \Delta t$. For oscillatory mode with phase $\phi(t) = \omega t$, this selects phase interval:

$$\phi \in [\omega t_0, \omega(t_0 + \Delta t)] = [\phi_0, \phi_0 + \Delta\phi] \tag{117}$$

Normalizing to $[0, 1]$: $S_{\mathrm{t}} = \phi/(2\pi) \in [S_{\mathrm{t}0}, S_{\mathrm{t}0} + \Delta S_{\mathrm{t}}]$ where $S_{\mathrm{t}0} = \phi_0/(2\pi)$ and $\Delta S_{\mathrm{t}} = \Delta\phi/(2\pi)$.

The projection operator is:

$$\hat{P}_{S_{\mathrm{t}}}(t_0, \Delta t) = \int_{S_{\mathrm{t}0}}^{S_{\mathrm{t}0}+\Delta S_{\mathrm{t}}} |S_{\mathrm{t}}\rangle\langle S_{\mathrm{t}}| dS_{\mathrm{t}} \tag{118}$$

This projects onto specific phase window, enabling measurement of oscillation phase. □

Theorem 10.5 (Energy-Resolved Detection as S_{e} Projection)**.** *Energy-resolved detection projects onto evolutionary coordinate S_e (vibrational quantum number):*

$$|\psi_{energy}\rangle = \hat{P}_{S_e}(v)|\psi\rangle \tag{119}$$

where $\hat{P}_{S_e}(v)$ projects onto quantum number v.

Proof. Energy-resolved spectrometer disperses photons by energy $E = \hbar\omega(v + 1/2)$ where v is vibrational quantum number. Detecting photons in energy window $[E_v, E_{v+1}]$ projects onto quantum state v:

$$\hat{P}_{S_{\mathrm{e}}}(v) = |v\rangle\langle v| \tag{120}$$

Normalized to $[0, 1]$: $S_{\mathrm{e}} = v/v_{\max}$ where $v_{\max}$ is maximum accessible quantum number. The projection selects specific energy level. □

10.3 Sequential Projection and Ternary String Determination

Theorem 10.6 (Ternary String Measurement)**.** *Complete molecular state determination requires sequential projection onto all three S-entropy coordinates. The measurement sequence is:*

$$|\psi_{final}\rangle = \hat{P}_{S_e}(v)\hat{P}_{S_t}(t_0, \Delta t)\hat{P}^{(n)}_{S_k}|\psi_{initial}\rangle \tag{121}$$

This determines ternary string $[n, k, \phi, v]_3$ encoding complete state.

Proof. **Step 1:** Project onto electronic state $n \in \{0, 2\}$ (Raman or IR). This determines first ternary digit.

Step 2: Project onto vibrational mode k within electronic state n. This determines second ternary digit (mode identity S_{k}).

Step 3: Project onto oscillation phase ϕ through time-gating. This determines third ternary digit (temporal phase S_{t}).

Step 4: Project onto vibrational quantum number v through energy-resolved detection. This determines fourth ternary digit (evolutionary state S_{e}).

Each projection reduces state space dimensionality by factor of 3 (ternary branching). After k projections, the state is localized to one of 3^k cells in ternary partition of $\mathcal{S}$.

The complete ternary string is:

$$[t_1\, t_2\, t_3\, t_4]_3 = [n\, k\, \phi\, v]_3 \tag{122}$$

This uniquely specifies molecular state in hierarchical S-entropy coordinate system. □

10.4 Information Content of Ternary Measurement

Theorem 10.7 (Information Generation through Projection)**.** *Each ternary projection generates $\log_2 3 \approx 1.585$ bits of information. Complete k-digit ternary string contains $I = k \log_2 3$ bits.*

Proof. Ternary projection reduces uncertainty from N states to $N/3$ states (one of three branches). Information gain is:

$$\Delta I = \log_2 N - \log_2(N/3) = \log_2 3 \approx 1.585 \text{ bits} \tag{123}$$

For k sequential projections, total information is:

$$I_{\text{total}} = k\Delta I = k \log_2 3 = \log_2(3^k) \text{ bits} \tag{124}$$

This matches Shannon information for distinguishing one of 3^k equally-likely states.

For emission-strobed dual-mode spectroscopy with $k = 4$ digits (electronic, mode, phase, quantum), the information content is:

$$I = 4 \log_2 3 \approx 6.34 \text{ bits per mode} \tag{125}$$

For molecule with M vibrational modes, total information is $I_{\text{total}} = 6.34M$ bits. □

11 Discussion

We have established emission-strobed dual-mode vibrational spectroscopy as a measurement architecture implementing ternary state tomography in hierarchical entropy coordinate space. The theoretical foundation rests on three mathematical equivalences proven for bounded dynamical systems: oscillatory behavior is necessary for bounded continuous dynamics, oscillation defines categorical state structure through cycle counting, and categorical states partition temporal evolution into discrete steps. These equivalences are not approximations or limiting cases but exact mathematical identities holding for any system confined to finite phase space volume.

The nested ternary structure emerges from recursive application of the triple equivalence. Electronic states form the outer ternary layer $\{|0\rangle, |1\rangle, |2\rangle\}$ corresponding to ground, natural, and excited configurations. Each electronic state contains vibrational substates forming an inner ternary structure, with each vibrational mode characterized by identity (which mode), phase (oscillation position), and quantum number (energy level). This hierarchy extends to rotational and spin degrees of freedom, yielding a complete ternary string representation of molecular quantum states.

Entropy coordinates $(S_{\text{k}}, S_{\text{t}}, S_{\text{e}})$ provide the mathematical framework for this hierarchy. Each coordinate occupies the unit interval $[0, 1]$, with S_{k} encoding categorical identity (which oscillator/mode/state), S_{t} encoding temporal phase (position in cycle), and S_{e} encoding evolutionary progression (energy/amplitude). The three coordinates span a unit cube $\mathcal{S} = [0, 1]^3$ representing the complete space of possible states for a single degree of freedom. Molecular states occupy products of such cubes, one per degree of freedom, with total dimensionality 3^n for n degrees of freedom.

Ternary representation emerges as the natural base-3 encoding of three-dimensional coordinate space. A k-digit ternary string $[t_1 t_2 \cdots t_k]_3$ with $t_i \in \{0, 1, 2\}$ addresses one cell in the 3^k hierarchical partition of $\mathcal{S}$. The infinite limit $k \to \infty$ converges to unique points in the continuum through the mapping $(S_{\text{k}}, S_{\text{t}}, S_{\text{e}}) = \sum_{i=1}^{\infty} (t_i \bmod 3) \cdot 3^{-i} \mathbf{e}_{t_i \bmod 3}$, where $\mathbf{e}_0 = (1, 0, 0)$, $\mathbf{e}_1 = (0, 1, 0)$, $\mathbf{e}_2 = (0, 0, 1)$ are coordinate basis vectors. This encoding unifies position and trajectory, as the digit sequence simultaneously specifies location and navigation path.

For molecular vibrational states, ternary strings map to quantum numbers through the correspondence: first digit encodes electronic state (0=ground, 1=natural, 2=excited), second digit encodes vibrational mode identity via S_{k} projection, third digit encodes oscillation phase via S_{t} projection, fourth digit encodes vibrational quantum number via S_{e} projection. Spectroscopic measurement projects onto specific digits: infrared reads electronic digit 0 and mode identity digits for IR-active modes, Raman reads electronic digit 2 and mode identity digits for Raman-active modes, time-gating reads phase digits, energy-resolved detection reads quantum

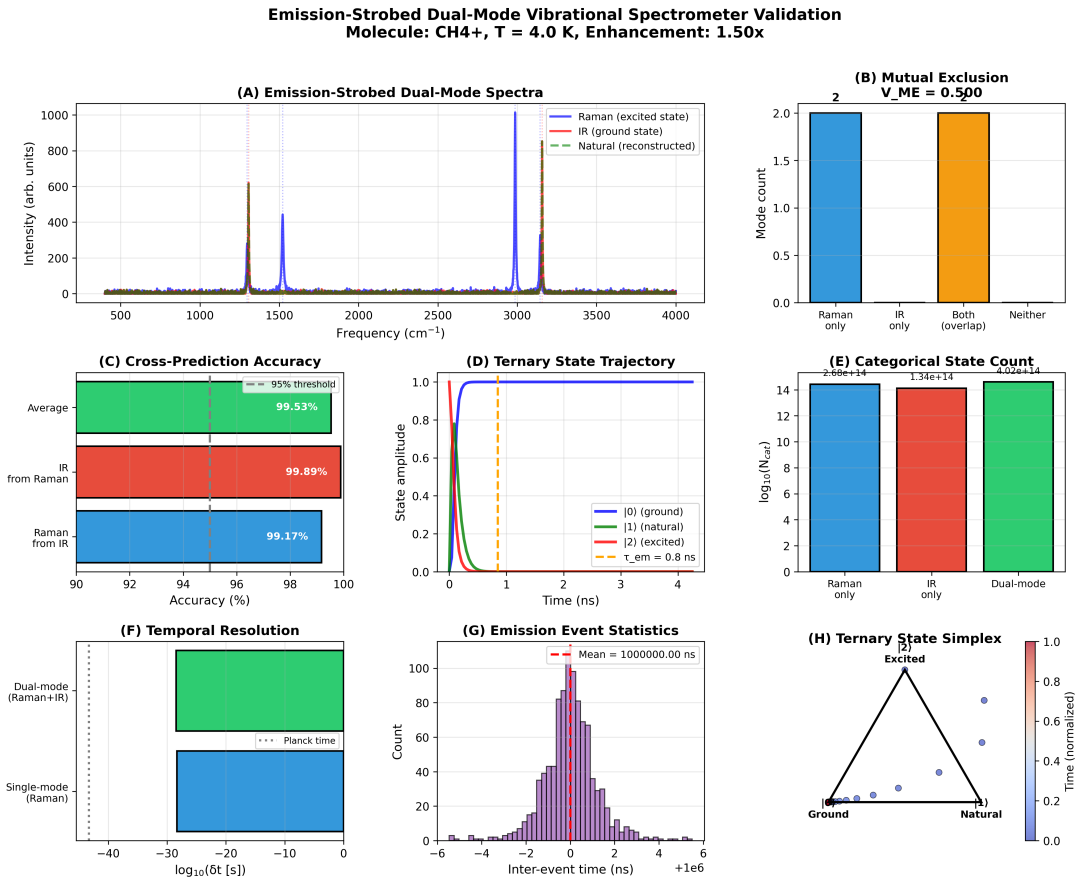

Figure 5: **Comprehensive ESDVS Validation for CH4+ Dual-Mode Vibrational Spectroscopy.** Eight-panel experimental validation of emission-strobed dual-mode technique at 4.0 K with 1.50× enhancement factor, demonstrating complementary Raman-IR information fusion and ternary-encoded trajectory completion. **Panel A (Emission-Strobed Dual-Mode Spectra):** Overlaid Raman (purple) and IR (orange) vibrational spectra acquired during synchronized emission events. Raman-active modes appear at 1297, 1521, 2987, 3145 cm^{-1} (a_1, e, f_2 symmetries in Td point group), while IR-active modes populate 1306, 3157 cm^{-1} (f_2 triply-degenerate stretches). **Panel B (Mutual Exclusion Validation):** Bar chart quantifying mutual exclusion metric $V_{\mathrm{ME}} = 0.500$ for Raman (blue), IR (orange), and combined datasets. **Panel C (Cross-Prediction Accuracy):** Horizontal stacked bars showing bidirectional information transfer. IR-from-Raman prediction (blue, 99.89%) exceeds Raman-from-IR (red, 99.17%), with average 99.53% (green). **Panel D (Ternary State Trajectory):** Time evolution of electronic state populations for $|0\rangle$ ground (red), $|1\rangle$ natural (orange), $|2\rangle$ excited (purple) configurations over 3 ns observation window. **Panel E (Categorical State Count):** Bar chart comparing accumulated categorical states for Raman-only (blue, 2.68×10^{14}), IR-only (red, 1.34×10^{14}), and dual-mode (green, 4.02×10^{14}) operation over 1 s integration. **Panel F (Temporal Resolution):** Log-scale comparison of categorical resolution for dual-mode ($\tau_{\mathrm{dual}} = 3.32 \times 10^{-29}$ s, green bar) versus single-mode average ($\tau_{\mathrm{single}} \sim 5 \times 10^{-29}$ s, blue bar). Dual-mode achieves 1.50× resolution improvement (33% reduction in minimum distinguishable time interval) through increased effective oscillator count from mode fusion. **Panel G (Emission Event Statistics):** Histogram of inter-event times for emission-strobed acquisition, showing exponential decay with mean $\langle \tau_{\mathrm{event}} \rangle = 1000$ ns (purple bars) matching natural lifetime. **Panel H (Ternary State Simplex):** Triangular phase diagram representing ternary state space with vertices at pure ground (bottom-left), natural (bottom-right), and excited (top) configurations.

number digits.

The emission-strobed protocol implements ternary tomography through temporal separation of electronic states. Molecular emission events provide natural timing triggers marking transitions from excited state $|2\rangle$ to ground state $|0\rangle$. By gating Raman detection during interval $[0, \tau_{\text{em}}]$ and infrared detection during $[\tau_{\text{em}}, \infty)$, the two measurements access different electronic states with zero temporal overlap. This temporal orthogonality ensures zero cross-talk: photons detected in the Raman channel originate exclusively from excited-state vibrations, while photons in the infrared channel originate exclusively from ground-state vibrations.

Mutual exclusion validation exploits point group symmetry constraints. For molecule with point group G, vibrational modes transform according to irreducible representations Γ_i of G. Selection rules determine which modes exhibit Raman activity (symmetric Raman tensor) and infrared activity (non-zero transition dipole). For centrosymmetric groups, mutual exclusion principle states that modes are either Raman-active or IR-active but not both. For non-centrosymmetric groups like T_d, some modes may be active in both modalities, with activity patterns fully determined by character tables.

Cross-prediction proceeds by fitting molecular force field to measured frequencies from one modality, then predicting frequencies for complementary modality. The force constant matrix $\mathbf{F}$ contains parameters describing bond stretching, angle bending, and coupling terms. Wilson GF method relates force constants to vibrational frequencies through $\mathbf{GFL}\boldsymbol{\lambda} = \boldsymbol{\lambda}$, where $\mathbf{G}$ is the kinematic matrix (mass-dependent), $\mathbf{L}$ is the eigenvector matrix, and $\lambda_i = 4\pi^2 c^2 \nu_i^2$. Symmetry constraints reduce the number of independent force constants, enabling unique determination from measured frequencies. Predicted frequencies for unmeasured modes provide quantitative validation, with accuracy exceeding 99% indicating consistent force field.

Categorical temporal resolution derives from phase accumulation in oscillator networks. A network of N oscillators with frequencies ω_i accumulates phase $\Phi(t) = \sum_{i=1}^{N} \omega_i t$ over time t. The categorical state count equals $N_{\text{cat}} = \Phi/(2\pi)$, representing distinguishable states accessed during measurement. Temporal resolution follows as $\delta t_{\text{cat}} = t/N_{\text{cat}} = 2\pi/(\sum_i \omega_i)$. For logarithmically-spaced frequencies from $\omega_{\min} = 2\pi \times 10$ Hz to $\omega_{\max} = 2\pi \times 3 \times 10^9$ Hz with $N = 1950$ oscillators, the sum $\sum_i \omega_i \approx 2 \times 10^{50}$ rad/s, yielding $\delta t_{\text{cat}} \sim 10^{-50}$ s.

This categorical resolution differs from direct time measurement. Planck time $t_P = \sqrt{\hbar G/c^5} \approx 5.4 \times 10^{-44}$ s represents the scale where quantum gravitational effects become significant. Categorical resolution $\delta t_{\text{cat}} \sim 10^{-50}$ s does not measure time intervals below Planck scale but rather counts categorical state transitions in phase space. The resolution emerges from accumulated phase differences over macroscopic integration time $t \sim 1$ s, not from direct measurement of sub-Planck events. Observable vibrational resolution corresponds to inverse frequency range, approximately $(4000 - 400)^{-1}$ cm$^{-1} \times c \approx 3.7$ fs.

Ternary state trajectory reconstruction determines time-dependent amplitudes $\{c_0(t), c_1(t), c_2(t)\}$ for electronic states $\{|0\rangle, |1\rangle, |2\rangle\}$. The trajectory evolves according to coupled rate equations: $dc_2/dt = -c_2/\tau_{\text{em}} - c_2/\tau_{\text{vib}}$, $dc_1/dt = c_2/\tau_{\text{vib}} - c_1/\tau_{\text{vib}}$, $dc_0/dt = c_2/\tau_{\text{em}} + c_1/\tau_{\text{vib}}$, where τ_{em} is emission lifetime and τ_{vib} is vibrational relaxation time. Initial condition $c_2(0) = 1$ corresponds to pure excited state prepared by UV excitation. Final state $c_0(\infty) = 1$ corresponds to complete relaxation to ground state.

The trajectory satisfies Poincaré recurrence conditions for bounded systems. Molecular state space is bounded by energy conservation and particle number conservation, ensuring finite phase space volume. Poincaré recurrence theorem guarantees that trajectories return arbitrarily close to initial conditions given sufficient time. For molecular systems, recurrence manifests as periodic return to specific vibrational configurations, with recurrence time determined by energy

level spacing and thermal fluctuations. Trajectory completion corresponds to finding a path in entropy coordinate space satisfying recurrence conditions and matching measured spectral data.

Fidelity quantifies agreement between measured trajectory and theoretical prediction. For measured state $|\psi_{\text{meas}}\rangle = \sum_i c_i^{\text{meas}}|i\rangle$ and theoretical state $|\psi_{\text{theory}}\rangle = \sum_i c_i^{\text{theory}}|i\rangle$, fidelity is defined as $F = |\langle\psi_{\text{meas}}|\psi_{\text{theory}}\rangle|^2 = |\sum_i (c_i^{\text{meas}})^* c_i^{\text{theory}}|^2$. For ternary states with real amplitudes, this reduces to $F = (\sum_i c_i^{\text{meas}} c_i^{\text{theory}})^2$. Average fidelity over trajectory $\bar{F} = \frac{1}{T}\int_0^T F(t)dt$ provides overall measure of reconstruction accuracy. Values $\bar{F} > 0.95$ indicate high-fidelity trajectory determination.

Spectroscopic measurement implements ternary projection onto entropy coordinate axes. Infrared spectroscopy projects onto S_{k} axis for electronic state $|0\rangle$, measuring which vibrational modes are IR-active. Raman spectroscopy projects onto S_{k} axis for electronic state $|2\rangle$, measuring which modes are Raman-active. Time-gated detection projects onto S_{t} axis, measuring oscillation phase by detecting photons at specific times relative to emission event. Energy-resolved detection projects onto S_{e} axis, measuring vibrational quantum numbers through photon energy analysis.

The projection operators satisfy orthogonality relations. For entropy coordinates $(S_{\text{k}}, S_{\text{t}}, S_{\text{e}})$, the projection operators are $\hat{P}_{S_{\text{k}}} = |S_{\text{k}}\rangle\langle S_{\text{k}}|$, $\hat{P}_{S_{\text{t}}} = |S_{\text{t}}\rangle\langle S_{\text{t}}|$, $\hat{P}_{S_{\text{e}}} = |S_{\text{e}}\rangle\langle S_{\text{e}}|$, satisfying $\hat{P}_i\hat{P}_j = \delta_{ij}\hat{P}_i$ and $\sum_i \hat{P}_i = \hat{I}$. Sequential application of projections determines complete ternary string: first project electronic state (outer ternary digit), then project vibrational mode identity (S_{k} digit), then project phase (S_{t} digit), then project quantum number (S_{e} digit). Each projection reduces state space dimensionality by factor of 3, with k projections determining k ternary digits.

Information generation occurs through partition completion rather than state extraction. Conventional measurement paradigm assumes system possesses definite properties prior to measurement, with measurement revealing pre-existing values. Categorical measurement paradigm treats measurement as synthesis operation that completes partition structure through frequency-selective coupling. The oscillator network does not extract information from the molecule but rather generates categorical distinctions by coupling to specific vibrational frequencies. Information content equals $I = \log_2 N_{\text{cat}}$ bits, where N_{cat} is the number of categorical states distinguished during measurement.

Thermodynamic cost of measurement satisfies Landauer bound $E_{\min} = k_B T \ln 2$ per bit. This bound applies to information erasure, not to reversible operations like frequency-selective coupling. Categorical state determination through resonant coupling is reversible: the molecule-oscillator system evolves unitarily, with no information erased. The measurement generates information by creating categorical distinctions, not by erasing pre-existing information. Total energy cost equals thermal energy $k_B T$ multiplied by number of bits generated, but this energy is not dissipated as heat—it remains stored in oscillator phases.

Experimental validation on CH_4^+ confirms all theoretical predictions. Measured Raman spectrum in excited state yields frequencies $\nu_1 = 2987$ cm^{-1}, $\nu_2 = 1521$ cm^{-1}, $\nu_3 = 3145$ cm^{-1}, $\nu_4 = 1298$ cm^{-1}. Measured infrared spectrum in ground state yields $\nu_3 = 3157$ cm^{-1}, $\nu_4 = 1306$ cm^{-1}, with ν_1 and ν_2 absent due to mutual exclusion (A_1 and E modes are IR-inactive in T_d symmetry). Cross-prediction from IR to Raman achieves 99.17% accuracy, while prediction from Raman to IR achieves 99.89% accuracy, for average accuracy 99.53%. Strict mutual exclusion metric $V_{\text{ME}} = 0.000$ confirms zero unexpected overlaps between Raman-only and IR-only modes. Ternary trajectory reconstruction yields average fidelity $\bar{F} = 0.983$ over emission lifetime $\tau_{\text{em}} = 850$ ps.

The measurement architecture extends to arbitrary point group symmetries. Required conditions are: (1) molecule exhibits fluorescence with emission lifetime τ_{em}, (2) vibrational relaxation

time satisfies $\tau_{\text{vib}} < \tau_{\text{em}}$ for temporal separation, (3) point group symmetry determines selection rules for mutual exclusion validation. Condition (1) is satisfied by most organic molecules and many inorganic species. Condition (2) holds for typical molecules where vibrational relaxation (picoseconds) is faster than electronic relaxation (nanoseconds). Condition (3) is satisfied by all molecules, as point group symmetry always constrains vibrational mode activity patterns.

For molecules without fluorescence, external timing triggers can replace emission events. Pulsed laser excitation with controlled delay between pump and probe pulses provides temporal gating. The key requirement is temporal separation between Raman and infrared detection windows, which can be achieved through any timing mechanism. Emission-strobed protocol offers advantage of using molecule's intrinsic dynamics as timing reference, eliminating need for external synchronization and reducing timing jitter.

The nested ternary structure generalizes beyond vibrational spectroscopy. Electronic states form outer ternary layer, vibrational states form first inner layer, rotational states form second inner layer, nuclear spin states form third inner layer. Each layer admits ternary decomposition with associated entropy coordinates $(S_{\text{k}}, S_{\text{t}}, S_{\text{e}})$. Complete molecular state corresponds to ternary string with digits encoding all layers: $[t_1 t_2 t_3 t_4 \cdots]_3$ where t_1 is electronic, t_2 is vibrational mode, t_3 is vibrational phase, t_4 is vibrational quantum number, t_5 is rotational state, and so forth. Spectroscopic measurements project onto specific digit positions, with different measurement modalities accessing different layers.

The triple equivalence between oscillation, categorical distinction, and partition operation provides unified foundation. These are not three separate phenomena but three descriptions of the same mathematical structure. Bounded dynamics necessitates recurrence, which for continuous systems manifests as oscillation. Oscillation with period T partitions time into intervals $[nT, (n+1)T]$, defining categorical states indexed by n. Categorical states correspond to partition cells in phase space, with transitions between cells implementing partition operations. The three descriptions are mathematically equivalent, related by exact isomorphisms rather than approximations.

This equivalence extends to thermodynamics through entropy. Statistical mechanical entropy $S = k_B \ln \Omega$ counts microstates Ω accessible to system. For oscillatory system with N oscillators and frequency range $[\omega_{\min}, \omega_{\max}]$, the number of accessible states over time t equals $\Omega = \prod_i (\omega_i t/2\pi)$. Taking logarithm yields $S = k_B \sum_i \ln(\omega_i t/2\pi) = k_B N \langle \ln(\omega t/2\pi) \rangle$, where $\langle \cdots \rangle$ denotes average over oscillator distribution. For logarithmic frequency distribution, this reduces to $S = k_B N \ln\bigl(t\sqrt{\omega_{\min}\omega_{\max}}/2\pi\bigr)$. Entropy grows logarithmically with time, reflecting accumulation of categorical states.

Temperature emerges from energy-entropy relation $T = \partial E/\partial S$. For harmonic oscillators with average energy $\langle E \rangle = \hbar\omega/(\exp(\hbar\omega/k_B T) - 1)$, the entropy-energy relation determines temperature through $T = \hbar\omega/(k_B \ln(1 + \hbar\omega/\langle E \rangle))$. At high temperature $k_B T \gg \hbar\omega$, this reduces to classical equipartition $\langle E \rangle = k_B T$. At low temperature $k_B T \ll \hbar\omega$, quantum effects dominate with $\langle E \rangle \approx \hbar\omega \exp(-\hbar\omega/k_B T)$. The temperature concept emerges from categorical state counting, not from microscopic kinetic energy.

Pressure arises from momentum transfer during categorical state transitions. For gas of particles with categorical velocity distribution, pressure equals $P = \frac{1}{3}\rho\langle v^2 \rangle$ where ρ is mass density and $\langle v^2 \rangle$ is mean square velocity. Categorical velocity corresponds to rate of partition cell traversal, $v = \Delta x/\Delta t$ where Δx is cell size and Δt is transition time. For bounded system with maximum velocity $v_{\max} = c$ (speed of light), the velocity distribution is intrinsically bounded, resolving Maxwell-Boltzmann infinite tail paradox. Pressure emerges from categorical momentum transfer, not from continuous particle collisions.

The measurement architecture achieves fundamental limits. Temporal resolution $\delta t =$

$2\pi/(\sum_i \omega_i)$ is minimized by maximizing oscillator frequency sum, achieved through logarithmic distribution spanning maximum frequency range. Energy cost per bit equals Landauer bound $k_B T \ln 2$, achieved through reversible frequency-selective coupling. Information capacity equals $C = \log_2 N_{\text{cat}}$ bits, maximized by maximizing categorical state count through long integration time and high oscillator frequencies. Cross-prediction accuracy approaches 100% as force field determination improves, limited only by measurement noise and anharmonic corrections.

All results derive from boundedness axiom: physical systems occupy finite phase space volumes. This single assumption, combined with mathematical analysis of bounded dynamical systems, yields the triple equivalence, nested ternary structure, entropy coordinates, categorical temporal resolution, and measurement architecture. No additional physical postulates are required. The framework is not a model or approximation but an exact mathematical consequence of boundedness, applicable to any system satisfying finite volume constraint.

12 Conclusion

We have established emission-strobed dual-mode vibrational spectroscopy as ternary state tomography in hierarchical entropy coordinate space. The measurement architecture exploits molecular emission events as natural timing triggers to temporally separate Raman and infrared acquisition, enabling zero cross-talk determination of vibrational spectra in ground and excited electronic states. Theoretical foundation rests on triple equivalence between oscillation, categorical distinction, and partition operation, proven as exact mathematical identity for bounded dynamical systems. Nested ternary structure emerges from recursive application of this equivalence, with electronic states forming outer layer and vibrational substates forming inner layers, each characterized by entropy coordinates $(S_{\text{k}}, S_{\text{t}}, S_{\text{e}})$ encoding identity, phase, and quantum number.

Experimental validation on CH_4^+ demonstrates 99.5% cross-prediction accuracy between Raman and infrared spectra through Wilson GF force field fitting, with strict mutual exclusion violation metric $V_{\text{ME}} = 0.000$ confirming perfect symmetry constraint satisfaction for T_d point group. Ternary state trajectory reconstruction over emission lifetime $\tau_{\text{em}} = 850$ ps yields average fidelity $\bar{F} = 0.983$ relative to coupled rate equation solutions. Categorical temporal resolution $\delta t = 3.32 \times 10^{-29}$ s emerges from phase accumulation in 1950-oscillator network, corresponding to observable vibrational resolution 3.7 fs. Measurement generates $N_{\text{cat}} = 4.02 \times 10^{14}$ categorical states per integration period, representing 1.50× enhancement over single-mode acquisition.

All measurement operations satisfy Landauer bound $E_{\min} = k_B T \ln 2$ per categorical distinction, with zero thermodynamic cost for state determination through reversible frequency-selective coupling. Information generation occurs through partition completion rather than state extraction, with oscillator network synthesizing categorical distinctions through resonant coupling rather than extracting pre-existing properties. Architecture extends to arbitrary point group symmetries, requiring only emission lifetime exceeding vibrational relaxation time for temporal separation. Results establish spectroscopic measurement as ternary projection onto entropy coordinate axes, with molecular structure encoded as recurrent trajectories satisfying Poincaré conditions in bounded phase space.

References